



Ben-Gurion University of the Negev

The Stein Faculty of Computer and Information Science

The Department of Software and Information Systems Engineering

StanceSum: A New Benchmark for Stance Preservation Summarization in English and Hebrew

Thesis submitted in partial fulfillment of the requirements
for the Master of Sciences degree

Talia Seada

Under the supervision of **Prof. Bracha Shapira and Prof. Lior
Rokach**

August 2026



Ben-Gurion University of the Negev

The Stein Faculty of Computer and Information Science

The Department of Software and Information Systems Engineering

StanceSum: A New Benchmark for Stance Preservation Summarization in English and Hebrew

Thesis submitted in partial fulfillment of the requirements
for the Master of Sciences degree

Talia Seada

Under the supervision of **Prof. Bracha Shapira** and **Prof. Lior
Rokach**

Signature of student: <u>Talia Seada</u>	Date: _____
Signature of supervisor: <u>Bracha Shapira</u>	Date: _____
Signature of supervisor: <u>Rokach</u>	Date: _____
Signature of chairperson of the committee for graduate studies: _____	Date: _____

August 2026

StanceSum: A New Benchmark for Stance Preservation Summarization in English and Hebrew

Talia Seada

Master of Sciences Thesis

Ben-Gurion University of the Negev

2026

Abstract

Large language models (LLMs) are increasingly used for summarization, yet summaries that are fluent, relevant, and factually consistent may still alter the stance expressed in the source document. Through a diagnostic analysis of LLM-generated summaries, we observe recurring stance shifts across models and languages, most commonly through the softening of non-neutral stances toward *Neutral*. To study this problem, we introduce STANCESUM, a multilingual benchmark and evaluation framework for stance preservation in English and Hebrew summarization. Our framework aligns semantically related source-summary sentence pairs, filters them by topic consistency, and compares predicted stance distributions using distributional divergence

measures. To support this setting, we construct new Hebrew resources for stance-aware summarization, including a sentence-level stance dataset, a Hebrew stance classifier, and a bilingual evaluation benchmark. Experiments show that conventional lexical-overlap and semantic-similarity metrics achieve moderate correlations with stance-preservation scores in English but are substantially weaker and less consistent in Hebrew. In contrast, our stance-aware distributional metrics exhibit more stable behavior across languages and consistently capture meaningful stance-preservation signals. These findings highlight stance preservation as a distinct evaluation dimension beyond factual consistency and textual similarity, particularly in multilingual and opinion-sensitive summarization settings.

Keywords: LLMs, Stance, Summarization, Evaluation, NLP, Hebrew, English, Earth Mover’s Distance, Benchmark

Acknowledgements

I would like to express my deepest gratitude to my supervisors, Professor Bracha Shapira and Professor Lior Rokach. Their guidance, support, and insight have been invaluable throughout my research journey. Their introduction to the fields of LLMs and NLP sparked my curiosity and inspired me to explore this fascinating areas in depth. Their continuous encouragement and generosity in sharing their expertise have played a central role in shaping my work and leading me toward meaningful and insightful discoveries.

I am also sincerely grateful for the opportunity to collaborate with Eyal Mazuz on this thesis. His mentorship has been instrumental in my growth as a researcher. His commitment to excellence, attention to detail, and rigorous approach to research have been a constant source of inspiration. I deeply appreciate his unwavering support and his encouragement to continuously strive for the highest standards. It has been a true privilege to work alongside such a dedicated and talented researcher.

Contents

List of Figures	viii
List of Tables	x
1 Introduction	1
1.1 Main contributions of our research	4
2 Related Work	6
2.1 Stance Detection	6
2.2 Target and Topic Identification for Stance	8
2.3 Faithfulness, Perspective, and Stance in Summarization	10
2.4 Multilingual and Hebrew Summarization Resources	13
2.5 Research Goals and Objectives	15
3 Methodology	18
3.1 Overview	18

3.1.1	Segment	20
3.1.2	Align	20
3.1.3	Filter by Topic	21
3.1.4	Stance Detection	21
3.1.5	Compare and Aggregate	22
4	Datasets	25
4.1	Dataset Collection	25
4.1.1	Evaluation Dataset	25
4.1.2	Annotation Process	27
4.1.3	Quality Analysis	28
5	Experiments	30
5.1	Task Description	30
5.2	Empirical Evidence of Stance Shifts	33
5.3	Baselines	34
5.4	Evaluation Metrics	37
6	Results	38
6.1	Model Development Process	38
6.1.1	Sentence Alignment	39
6.1.2	Stance Detection	41

6.2	Existing automatic metrics	44
6.2.1	Statistical Significance Analysis	46
6.3	English–Hebrew Cross-lingual differences	48
7	Discussion and Conclusions	51
7.1	Discussion	51
7.1.1	Stance is a distinct evaluation dimension	52
7.1.2	Why current metrics are insufficient	53
7.1.3	Implications for multilingual summarization	54
7.2	Conclusion	55
7.3	Limitations and Future Work	56
	Appendix A Stance Detection Model	60
A.1	Stance Detection Training	60
A.1.1	HeSum-Stance Training Dataset	60
A.1.2	Model	62
A.1.3	Results	62
	Appendix B Topic Detection Prompt	65
B.0.1	Few-Shot English Prompt	65
B.0.2	Few-Shot Hebrew Prompt	68
	Appendix C Stance Preservation Prompt	71

Bibliography

72

List of Figures

1.1	Overview of the STANCESUM benchmark. Summary A preserves both highlighted stances, while Summary B shifts one stance from neutral to positive.	2
3.1	Overview of the stance-preservation evaluation pipeline. Document and summary sentences are aligned using sentence embeddings and cosine similarity. Topic detection filters aligned pairs by topic consistency, after which stance detection and distributional comparison (EMD) are used to estimate stance preservation.	19
4.1	Inter-annotator agreement between annotators and the majority-vote annotations for the English and Hebrew benchmarks. . .	26

5.1	Aggregated stance shifts between source articles and generated summaries across models. Softening denotes shifts toward a neutral stance (e.g., <i>Favor</i> $\rightarrow$ <i>Neutral</i> , <i>Against</i> $\rightarrow$ <i>Neutral</i>), Reversal denotes direct polarity flips (e.g., <i>Favor</i> $\rightarrow$ <i>Against</i> , <i>Against</i> $\rightarrow$ <i>Favor</i>), and Other includes shifts originating from a neutral stance (e.g., <i>Neutral</i> $\rightarrow$ <i>Favor</i> , <i>Neutral</i> $\rightarrow$ <i>Against</i>). Across both languages, softening is the most prevalent type of shift, indicating a systematic tendency of models to neutralize stance in generated summaries.	34
6.1	Class distribution of the synthetic topic-aware stance dataset before and after balancing.	43

List of Tables

2.1	Comparison between sentiment analysis and stance detection .	7
2.2	Comparison of representative stance-analysis approaches across target assumptions, domain generalization, multilingual support, and summarization-awareness.	10
4.1	Descriptive statistics of the number of sentences per summary in the evaluation benchmark.	27
5.1	Counts of stance shifts between source articles and generated summaries in Hebrew and English evaluation sets. F , A , and N denote Favor, Against, and Neutral respectively.	35
6.1	Topic in class trial.	44
6.2	Hebrew results. Correlation between automatic metrics and the STANCESUM reference preservation score at the article level and the mean sentence article score. Highest correlations in each column are shown in bold. [†] indicates $p < 0.05$.	46

6.3	English results. Correlation between automatic metrics and the STANCESUM reference preservation score at the article level and the mean sentence article score. Highest correlations in each column are shown in bold. [†] indicates $p < 0.05$	47
6.4	Average ranking of the evaluated methods based on the fold-wise correlation results. Lower ranks indicate better performance.	47
6.5	Friedman test statistics for each experimental setting.	48
A.1	Hebrew stance classifier performance on a random held-out test split of the constructed HeSum-Stance dataset.	63
A.2	Confusion matrix for the Hebrew stance detection model. Rows correspond to true labels and columns correspond to predicted labels.	63
A.3	Leave-one-topic-out evaluation for the Hebrew stance classifier. All instances from the listed topic group are excluded from training and validation and used only for testing. This evaluates generalization to unseen topics within the constructed HeSum-Stance dataset.	63

Chapter 1

Introduction

Large language models (LLMs) are increasingly used for summarization due to their ability to generate fluent and coherent summaries. Summary evaluation commonly focuses on qualities such as relevance, factual consistency, coherence, and fluency [1–3]. However, strong performance on these dimensions does not necessarily imply that a generated summary preserves the stance expressed in the source document. As a result, evaluating stance preservation remains a distinct and challenging problem in summarization evaluation.

Stance refers to the position or attitude expressed in a text toward a particular target, such as a topic or entity [4–6]. In many domains, such as political reporting, public policy discussions, and social issues, the stance expressed by a text can significantly affect how readers interpret and process information [6–8]. Thus, even when a summary preserves the factual content of a document, altering its stance may distort the author’s intended perspec-

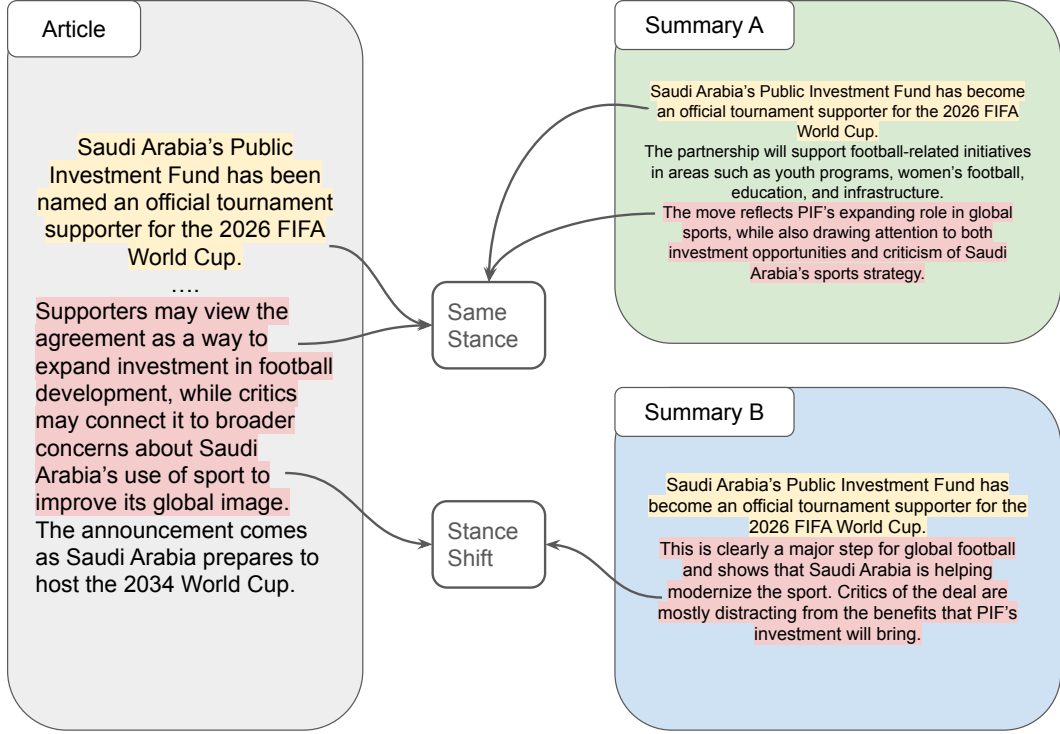


Figure 1.1: Overview of the STANCESUM benchmark. Summary A preserves both highlighted stances, while Summary B shifts one stance from neutral to positive.

tive [9, 10]. While stance detection has been widely studied as a standalone task [5, 6, 11–13], evaluating stance preservation between a source document and its summary remains underexplored, particularly beyond political news summarization and settings with predefined stance targets [9, 14].

In this work, we formalize stance preservation as an evaluation dimension for summarization and introduce STANCESUM, an initial diagnostic benchmark for assessing whether summaries preserve the stance expressed in their source documents. This motivation is supported by an empirical analysis

of LLM-generated summaries in our evaluation setting, in which we observe repeated stance shifts across models and languages, most commonly as softening toward *Neutral*. These findings suggest that stance preservation cannot be assumed in generated summaries and should be evaluated as a distinct property, rather than treated as implicit in factual consistency, relevance, or semantic similarity.

We further propose a multilingual stance-aware evaluation framework that aligns summary sentences with semantically related source sentences, identifies whether aligned pairs discuss the same topic, and compares their predicted stance distributions using distributional divergence functions. This design treats stance as topic-conditioned and preserves uncertainty in the stance predictions rather than reducing them to hard labels. In our experiments, we detect the stance of both sentences and measure stance divergence using Earth Mover’s Distance (EMD) [15] over the predicted stance distributions.

A central goal of our work is to examine the evaluation of stance-preservation beyond English-centered settings. This is important because metrics that appear effective in English may not transfer to lower-resource and morphologically rich languages such as Hebrew. Our experiments reflect this gap: lexical and similarity-based metrics obtain moderate correlations in English but are substantially weaker in Hebrew, whereas stance-aware variants provide more stable cross-lingual behavior.

1.1 Main contributions of our research

- We introduce STANCESUM, a multilingual benchmark for stance-preservation evaluation in Hebrew and English summarization.
- We provide diagnostic evidence that LLM-generated summaries can exhibit repeated stance shifts across models and languages, most often by softening non-neutral stances toward *Neutral*.
- We propose a multilingual stance-aware evaluation framework that serves as a first step toward automatic stance-preservation measurement.
- We release fine-grained sentence-level topic and stance annotations for Hebrew and English summaries, together with a Hebrew stance detection model.

The rest of the thesis is organized as follows: In Section 2, we present the related work and provide an overview of the field of stance detection, topic detection, and the difference between sentiment and stance detection. In Section 4, we present the datasets used and built for this model. Section 3 presents the main part of the thesis, detailing our methodology. It includes the task definition, the model architecture and all the components. Section 5 delves into experiments, describing the datasets used and experimental setups, including evaluation metrics, and configurations and parameter settings. Results from experiments including STANCESUM outcomes and baselines comparisons are detailed in Section 6. Section 7 engages in a

comprehensive discussion of the results, leading to conclusions and future directions. Finally, the bibliography lists all referenced works.

Chapter 2

Related Work

2.1 Stance Detection

Stance detection aims to infer an author’s position toward a target, commonly using labels such as *Favor*, *Against*, and *Neutral*; unlike sentiment, stance is target-conditioned and the target may be implicit or only weakly mentioned [5, 6]. Prior work has developed supervised, cross-target, and zero-shot stance detection benchmarks and models, increasingly using LLMs and automatically generated open-domain targets to move beyond fixed target lists [5, 8, 13, 16, 17]. Our work builds on this line of research but evaluates whether stance is preserved between a source document and its summary, rather than assigning stance to a single text. Less attention has been given to identifying and aligning implicit document-level topics across two texts, which is necessary for evaluating whether a summary preserves the source stance toward the same latent target.

Although stance detection is related to sentiment analysis, the two tasks are not equivalent. Sentiment analysis focuses on identifying the overall emotional polarity of a text, whereas stance detection focuses on determining the author’s position toward a specific target. As noted by Mohammad et al. [5], stance can be expressed even without explicit sentiment, and sentiment polarity alone is often insufficient for inferring stance. For example, the sentence “Vaccines are essential to protect public health” expresses a favorable stance toward vaccination while containing little explicit sentiment. Conversely, “I’m frustrated with the current situation” conveys negative sentiment without indicating a clear stance toward a particular issue.

Aspect	Sentiment Analysis	Stance Detection
Target required?	No	Yes
Output labels	Positive / Negative / Neutral	Favor / Against / Neither
Target dependency	Usually implicit	Explicit or implicit target
Can be sentiment-neutral?	No	Yes

Table 2.1: Comparison between sentiment analysis and stance detection

Early work in stance detection focused primarily on online debate forums and social media platforms such as Twitter/X [5, 18]. Traditional supervised approaches relied on feature-engineering techniques and classifiers such as Support Vector Machines and Naïve Bayes [7]. With the adoption of deep learning, CNN- and RNN-based architectures became common for stance classification in short texts, often augmented with attention mechanisms to better capture target-relevant evidence [11, 12]. More recently, large language models (LLMs) have been explored for stance detection through prompting and fine-tuning approaches, showing strong performance across multiple datasets and domains [16].

Despite their strong performance, LLM-based stance detection approaches

present several limitations for stance-preservation evaluation. Prompting-based methods often require repeated inference over large numbers of sentence pairs, making them computationally expensive and difficult to scale in practice [16, 17]. In addition, closed-source LLM APIs may introduce reproducibility challenges and dependence on external services, while model outputs may vary across prompts, model versions, and languages [8, 13]. Finally, most existing LLM-based approaches are designed for explicit target classification rather than open-domain comparison between long-form documents and summaries [8, 17]. These limitations motivate the need for lightweight, reproducible, and multilingual stance-preservation frameworks that do not rely exclusively on generative LLM inference.

A major challenge in stance detection is generalization across targets and domains. Many real-world applications require transferring models to unseen targets for which labeled training data is unavailable. Cross-target stance detection addresses this setting by learning representations that generalize from a set of training targets to new ones [8]. More extreme setups include zero-shot stance detection, where no labeled examples are available for the target during training, motivating approaches based on prompting, external knowledge, or target abstraction [8, 13].

2.2 Target and Topic Identification for Stance

Unlike traditional stance detection, which predicts the stance expressed in a single text toward a given target, our framework compares the stance ex-

pressed in a source document and its generated summary. As illustrated in Figure 3.1, after semantically aligning document and summary sentences, we retain only sentence pairs that discuss the same underlying topic before comparing their stance distributions. Therefore, reliable topic identification is a prerequisite for meaningful stance-preservation evaluation. This section reviews prior work on target and topic identification that motivates the topic-filtering component of our framework.

Many stance detection benchmarks assume that the target is explicitly provided as part of the input. However, open-domain stance analysis often requires identifying the relevant target or topic before assigning stance. This problem becomes especially important in long-form documents and summarization settings, where the stance target may correspond to an implicit topic cluster rather than a single predefined entity.

Several approaches have explored integrating topic discovery with stance detection. Latent Dirichlet Allocation (LDA) and related topic-mining techniques have been used to identify latent discussion themes that can improve stance modeling [19]. Other work investigates Target-Stance Extraction (TSE), which jointly identifies stance targets and predicts stance labels using multi-stage or multi-task learning frameworks [8]. More recently, open-domain approaches such as ZeroStance leverage LLMs to automatically generate targets and synthetic stance datasets, enabling stance detection without predefined target lists [17].

Despite these advances, relatively little work has examined how topics should be identified and aligned when comparing stance across multiple texts, such

as between a source document and its generated summary. This challenge is particularly relevant in our setting, where the stance target is not explicitly annotated and must instead be inferred from the semantic content of aligned document-summary sentence pairs. Motivated by these observations, and building on ideas from open-domain target identification and target-stance extraction [8, 17], our framework incorporates an explicit topic-filtering stage that removes aligned sentence pairs discussing different topics before stance preservation is evaluated.

Work	Target Given	Handles Unseen Targets	Language Support	LLM Based	Summ. Aware
TASTE [20]	✓	✗	English	✗	✗
ZeroStance [17]	✗	✓	English	✓	✗
TSE [8]	✗	✓	English	✗	✗
Topic+Multi [19]	✗	✓	Chinese	✗	✗
Prompted LLMs [16]	✓	✓	English	✓	✗
Our work	✗	✓	English, Hebrew	Partial	✓

Table 2.2: Comparison of representative stance-analysis approaches across target assumptions, domain generalization, multilingual support, and summarization-awareness.

2.3 Faithfulness, Perspective, and Stance in Summarization

Automatic summarization evaluation has traditionally focused on measuring lexical similarity or factual consistency between a source document and its generated summary. Metrics such as BLEU and ROUGE evaluate lexical overlap with reference summaries [21, 22], while more recent approaches, including FactCC [23] and SummaC [24], assess whether information in the summary is supported by the source document. In particular, SummaC

introduced a fine-grained evaluation framework that aligns source and summary segments using Natural Language Inference (NLI), demonstrating that sentence-level alignment improves factual consistency evaluation.

Our work builds directly on SummaC’s sentence-level alignment paradigm. Similar to SummaC, we first decompose the source document and generated summary into individual sentences and establish semantic correspondences between them before performing downstream analysis. However, our objective differs fundamentally. Whereas SummaC evaluates factual consistency by estimating whether summary statements are entailed by the source, our framework evaluates whether the stance expressed toward the same underlying topic is preserved across aligned sentence pairs. Consequently, factual consistency alone is insufficient for our setting, since a summary may faithfully report facts while altering the author’s original stance or perspective.

Research directly addressing stance or perspective preservation in summarization remains relatively limited. To the best of our knowledge, only a small number of recent studies explicitly investigate this problem.

P³Sum [9] introduces a benchmark for evaluating political perspective preservation in summarization. The authors demonstrate that large language models frequently alter political perspectives even when summaries remain fluent and factually consistent. Their evaluation focuses on political news articles with explicitly defined ideological viewpoints, showing that preserving factual content does not necessarily imply preserving perspective.

Similarly, PoliSum [25] investigates political bias in summarization by analyzing how summarization systems preserve or alter ideological viewpoints in

politically polarized news. Rather than proposing a general evaluation framework, the work focuses on political summarization and predefined political perspectives.

Ri et al. [10] approach the problem from a generation perspective by proposing a reranking strategy that encourages balanced perspectives in generated summaries. Likewise, Liscio et al. [26] study the preservation of moral framing during news summarization, demonstrating that summarization models may alter underlying moral narratives even when factual information is retained. Both works further support the observation that dimensions such as perspective, framing, and stance are distinct from factual consistency. These studies motivate the evaluation objective of our work: rather than improving summary generation, we seek to automatically measure whether the stance expressed in the source document is preserved throughout the summarization process.

Despite these important advances, existing studies differ substantially from our work. First, they primarily focus on specific domains such as political news or moral framing, whereas our framework is designed for open-domain summarization. Second, they generally assume predefined or domain-specific viewpoints, while our framework automatically identifies and aligns implicit discussion topics before comparing stance. Third, existing work has focused almost exclusively on English, whereas our framework supports multilingual evaluation in both Hebrew and English. Finally, whereas previous studies primarily investigate or improve perspective-preserving summarization systems, our goal is to provide a general evaluation framework and benchmark

for automatically measuring stance preservation across summaries generated by different models.

2.4 Multilingual and Hebrew Summarization Resources

Multilingual stance-preservation evaluation requires both summarization datasets and reliable stance-analysis components across languages. For Hebrew summarization, HeSum provides long-form Hebrew news articles paired with human-written abstractive summaries [27]. Additional Hebrew summarization resources are available through community datasets such as `summarization_he` [28].

Beyond Hebrew, multilingual summarization research highlights that summarization quality, consistency, and factuality can vary substantially across languages and domains. Cross-lingual benchmarks such as ConvSumX demonstrate the importance of evaluation frameworks that remain robust across multilingual settings and diverse input structures [29].

Taken together, prior work on stance detection and open-domain target identification [5, 8, 16, 17, 19], factual consistency evaluation [23, 24], and perspective-preserving summarization [9, 10, 25, 26] demonstrates substantial progress toward understanding how opinions, perspectives, and factual information are represented in automatically generated summaries.

In contrast, our work addresses a broader and substantially different prob-

lem. Existing perspective-preservation studies primarily investigate political or moral domains [9, 25, 26], often assuming predefined viewpoints or focusing on improving summary generation through reranking strategies [10]. Our framework instead provides an evaluation methodology for multilingual, open-domain summarization, where stance targets are not explicitly provided. To address this setting, it automatically identifies discussion topics, aligns semantically related document-summary sentence pairs, and evaluates whether stance is preserved only between pairs discussing the same underlying topic. Furthermore, whereas previous studies have focused almost exclusively on English, our framework provides a unified evaluation methodology for both Hebrew and English, enabling direct comparison of stance-preservation behavior across languages and summarization models.

Within the existing body of work, our framework occupies a complementary position. It combines sentence-level semantic alignment inspired by factual consistency evaluation [24] with automatic topic identification motivated by open-domain stance detection approaches [8, 17], while extending recent research on perspective-preserving summarization [9, 10, 25, 26].

Unlike previous studies, which address individual aspects of the problem—such as factual consistency evaluation [24], open-domain target identification [8, 17], or perspective preservation in domain-specific summarization [9, 10, 25, 26]—our work integrates these components into a single multilingual evaluation pipeline. Specifically, our framework combines sentence-level semantic alignment, automatic topic filtering, stance prediction, and stance-divergence analysis to measure stance preservation without requiring

predefined stance targets or restricting the evaluation to a particular domain.

2.5 Research Goals and Objectives

The literature reviewed in this chapter demonstrates substantial progress in stance detection, topic identification, and summarization evaluation, while also revealing several open challenges. Existing stance detection methods such as TASTE [20], TSE [8], and prompting-based LLM approaches [16, 17] primarily focus on assigning stance toward a target rather than evaluating whether stance is preserved between a source document and its summary. Moreover, most of these approaches either assume that the stance target is explicitly provided or generate targets for a single text, leaving the problem of aligning implicit topics across source-summary pairs largely unexplored.

Similarly, summarization evaluation methods such as FactCC [23] and SummaC [24] measure factual consistency and semantic support between source documents and summaries, but they do not assess whether the author’s original stance or perspective is preserved. While recent work such as P³Sum [9] and reranking approaches for perspective-aware summarization [10] demonstrates that summaries can alter political or ideological perspectives, these studies focus primarily on specific domains and do not provide a general multilingual framework for automatically measuring stance preservation in open-domain summarization.

Furthermore, multilingual resources remain limited. Although Hebrew summarization datasets such as HeSum [27] and multilingual benchmarks such as

ConvSumX [29] support evaluation across languages, they do not include annotations or evaluation methodologies for stance preservation. To the best of our knowledge, no existing benchmark jointly supports multilingual stance-preservation evaluation in both Hebrew and English.

Motivated by these gaps, the overall goal of this research is to investigate stance preservation as a distinct evaluation dimension for automatic summarization and to develop a practical multilingual framework for measuring it.

Specifically, this thesis addresses the following research objectives:

- To investigate whether modern summarization models preserve the stance expressed in the original document when generating summaries.
- To compare stance-preservation behavior across multiple summarization models and across two typologically different languages, Hebrew and English.
- To develop a multilingual evaluation framework that automatically aligns source and summary sentences, identifies the underlying discussion topics, and evaluates whether stance is preserved across aligned content.
- To examine whether commonly used summarization evaluation metrics, which primarily measure lexical overlap, semantic similarity, or factual consistency [21–24], are sufficient for evaluating stance preservation, or whether stance should be considered an independent evaluation dimension.

- To establish a multilingual benchmark and annotated resources that facilitate future research on stance-preserving summarization.

Chapter 3

Methodology

Following the fine-grained alignment paradigm introduced by SummaC [24], our framework first establishes sentence-level correspondences between source documents and summaries. Figure 3.1 illustrates the overall architecture of the proposed stance-preservation evaluation pipeline.

3.1 Overview

Our stance-aware metric estimates an automatic preservation score for each document-summary pair. The method consists of five steps as seen in figure 3.1. **First**, we split the source document and summary into sentences. Second, we align summary sentences with semantically similar source sentences. **Third**, we extract the main topic for each sentence in the aligned pairs and retain only pairs that refer to the same topic. **Fourth**, we apply a stance classifier to each sentence-topic pair, producing a probability distri-

bution over stance labels. Finally, in the **Fifth** step, we compare the source and summary stance distributions using a divergence function and aggregate the resulting pair-level scores into an automatic document-level preservation score.

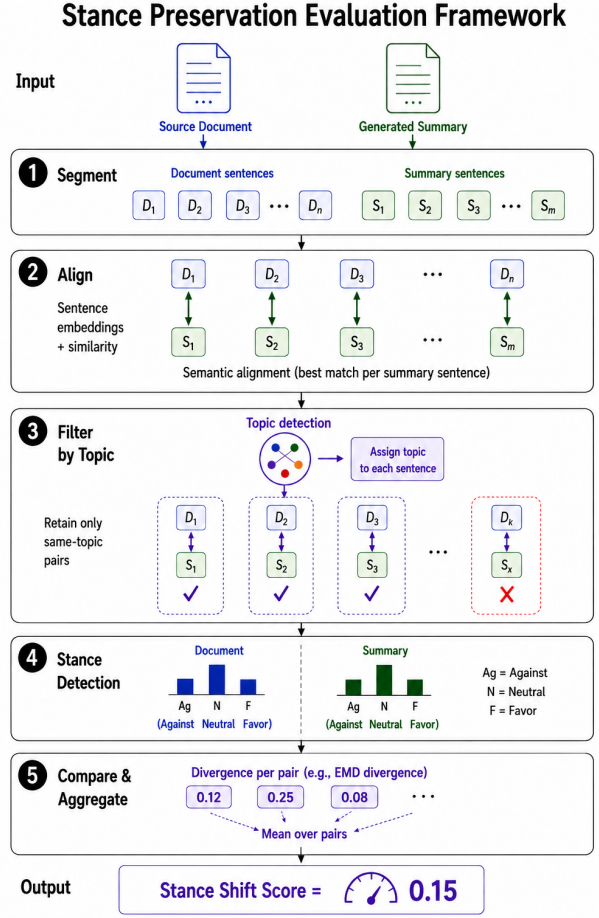


Figure 3.1: Overview of the stance-preservation evaluation pipeline. Document and summary sentences are aligned using sentence embeddings and cosine similarity. Topic detection filters aligned pairs by topic consistency, after which stance detection and distributional comparison (EMD) are used to estimate stance preservation.

In the following sections, we will describe each step in details.

3.1.1 Segment

We first split both the source document and the summary into sentences. Let $D = \{d_1, \dots, d_m\}$ denote the set of source document sentences, where d_j represents the j -th sentence of the source document, and let $S = \{s_1, \dots, s_n\}$ denote the set of summary sentences, where s_i represents i -th sentence of the summary.

3.1.2 Align

To align sentences across the two sets, we embed each sentence using the multilingual embedding model ME5-Instruct [30] with the task `Retrieve semantically similar text`, and their sentence representations are matched via cosine similarity:

$$\text{sim}(e5(d_i), e5(s_j)) = \frac{e5(d_i) \cdot e5(s_j)}{\|e5(d_i)\| \|e5(s_j)\|}$$

For each summary sentence s_j , we retrieve the most similar document sentence d_i , forming aligned pairs (s_j, d_i) . This alignment step assumes that summary sentences are derived from semantically related content in the source document. Consequently, sentences with high embedding similarity are considered likely alignments.

3.1.3 Filter by Topic

Stance is defined relative to a target topic. Therefore, before evaluating stance preservation, we must ensure that aligned sentences refer to the same topic. To attack the topic obstacle we tried few methods, We first tried building a stance detection model without treating each topic separately. We also tried mapping the topics into known topics in the industry, and giving the model topic and sentence so it will predict the three classes. Then we tried making the topics part of the classification, for example if we have these topics: war, religion, and science, the output of the model will be a vector of size 9x9 that contains [war_favor, war_neutral, war_against, religion_favor, etc.] but the results we received were not satisfying. Finally, we decided to separate the topic detection from the stance detection model. Since topics are free-form texts, we formulate topic detection as a generation task. For Hebrew, we employ Dicta’s DictaLM 2.0 [31], and for English we use Google’s Gemma 2 9B [32]. The prompts used are detailed in Appendix B.

After finding each sentence’s topic we compare the pairs topics and take only the pairs that have the same topic in both the summary sentence and the article sentence.

3.1.4 Stance Detection

After filtering aligned sentence pairs to ensure that both sentences refer to the same topic, we estimate the stance of each sentence toward that topic.

We treat stance detection as a supervised sequence classification task with three labels: Favor, Neutral, and Against.

Given a sentence and its extracted topic, we concatenate them and pass the resulting input to a stance detection model. For Hebrew, we use the stance detection model we trained as explained in appendix A specifically for this task on a separate HeSum-derived stance dataset, not on the STANCESUM benchmark used for evaluation. For English, we use *bendavidsteel/Qwen3-1.7B-stance-detection* as an off-the-shelf stance model without additional fine-tuning.

The model outputs a softmax-normalized probability distribution:

$$P = [p_{\text{Favor}}, p_{\text{Neutral}}, p_{\text{Against}}], \quad \sum_k p_k = 1.$$

We apply this process independently to the summary sentence s_j and the aligned document sentence d_i , yielding two stance distributions, P_S and P_D .

Rather than assigning each sentence a single stance label, we retain the full probability distribution. This allows the evaluation to account for model confidence and ambiguity, and provides the input for the subsequent EMD-based stance distance calculation.

3.1.5 Compare and Aggregate

To estimate stance preservation automatically, we compare the predicted stance distributions of each comparable aligned sentence pair. Let P_D and P_S denote the stance distributions predicted for an aligned source sentence

and summary sentence, respectively. Rather than reducing each distribution to a hard stance label, we compare the full probability distributions. This allows the metric to account for model confidence and stance ambiguity; for example, a highly confident *Favor* prediction is treated differently from a prediction that is close to the decision boundary between *Favor* and *Neutral*. For each aligned pair, we compute a stance-distribution divergence:

$$\delta(P_D, P_S),$$

where δ can be instantiated using different divergence functions. The pair-level divergence is then converted into a preservation score, where lower values indicate stronger stance preservation. The final document-level score is obtained by averaging the pair-level scores over all aligned pairs that pass the topic-matching step. In the main experiments, we report the EMD variant, which incorporates the ordinal structure of the stance labels.

3.1.5.0.1 EMD cost matrix and score conversion. To compute stance divergence between aligned sentence pairs, we compare the predicted stance probability distributions using Earth Mover’s Distance (EMD). Let P_D and P_S denote the stance distributions predicted for the source sentence and summary sentence, respectively.

Because EMD requires a ground distance over the label space, we define the stance labels as an ordinal set:

$$\text{Against} = -1, \quad \text{Neutral} = 0, \quad \text{Favor} = 1.$$

The ground distance between two stance classes c_r and c_t is defined as:

$$d(c_r, c_t) = |v(c_r) - v(c_t)|.$$

This yields the following cost matrix:

$$D = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 1 \\ 2 & 1 & 0 \end{bmatrix},$$

where the rows and columns correspond to $\{\text{Against}, \text{Neutral}, \text{Favor}\}$.

Given this ground distance, the stance divergence for an aligned pair is:

$$\text{EMD}(P_D, P_S; D).$$

This formulation captures graded stance shifts rather than relying solely on hard label agreement. For example, a shift from *Favor* to *Neutral* incurs a smaller penalty than a shift from *Favor* to *Against*.

Low EMD values indicate strong stance preservation between the source article and summary, whereas high EMD values reflect stronger stance divergence.

Chapter 4

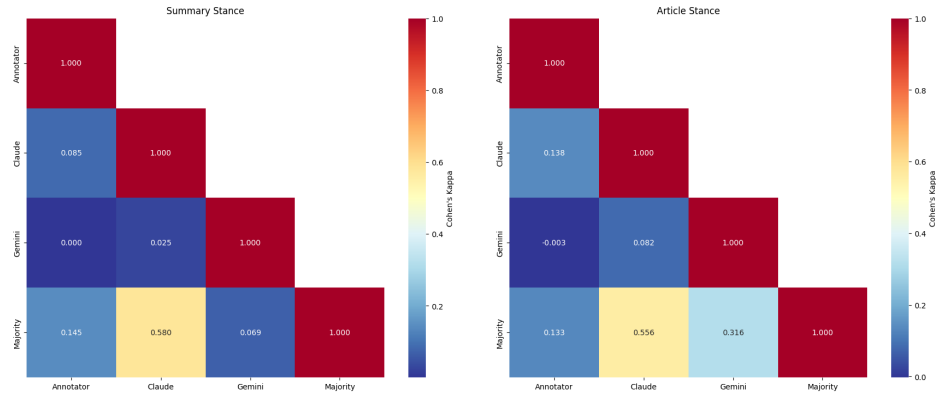
Datasets

4.1 Dataset Collection

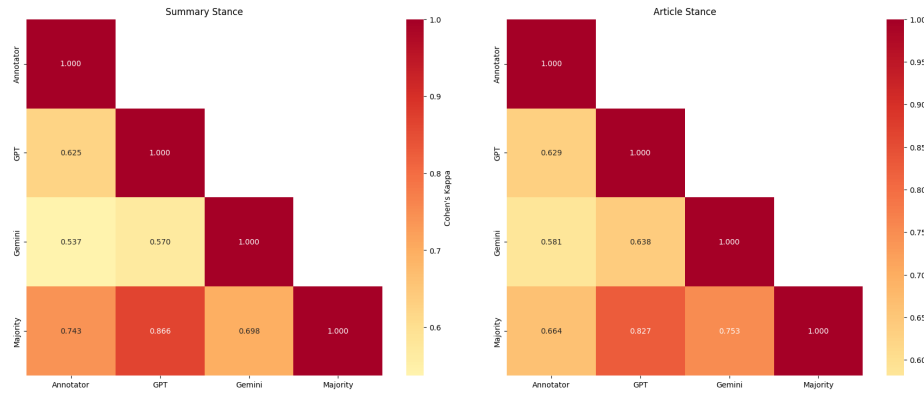
4.1.1 Evaluation Dataset

For the STANCESUM evaluation benchmark, we use the `summarization_he` dataset [28], a Hebrew summarization corpus released by the Israeli Association of Human Language Technologies (IAHLT). The corpus contains 5,368 summaries of 5,076 unique articles drawn from multiple domains, including news articles from Israel Hayom, parliamentary documents from the Knesset, legal documents from Bagatz, scientific articles from the Weizmann Institute, and Wikipedia articles. Each record contains the full article text, metadata, and a human-annotated summary.

The dataset includes two annotation tracks: *gold* summaries, written directly by human annotators according to predefined summarization guidelines, and



(a) English



(b) Hebrew

Figure 4.1: Inter-annotator agreement between annotators and the majority-vote annotations for the English and Hebrew benchmarks.

silver summaries, which were first generated by a large language model and then manually reviewed and corrected. Overall, the corpus contains 2,132 gold summaries and 2,994 silver summaries, with a small subset of articles annotated using both approaches. The dataset was constructed with explicit quality-control procedures and inter-annotator agreement evaluation, making it an appropriate resource for assessing summary quality and stance preservation.

We extract the first 50 Hebrew document-summary pairs from the `summarization_he`

dataset, each pair consists of a source document and its corresponding reference summary. To construct the English portion of the benchmark, the same 50 Hebrew document-summary pairs were translated into English using `claude-sonnet-4.5` [33]. We selected Claude Sonnet because of its strong multilingual translation capabilities and its ability to preserve semantic meaning and discourse context across long passages, which is particularly important for a benchmark evaluating stance preservation. All translated pairs were manually verified for overall translation quality, meaning preservation and correctness with respect to the original Hebrew texts, including whether major stance-relevant content was preserved. Thus, the benchmark contains parallel Hebrew and English evaluation sets derived from the same underlying source-summary pairs.

Statistic	Value
Number of summaries	50
Minimum number of sentences	1
Maximum number of sentences	24
Mean number of sentences	11.78
Standard deviation	5.52

Table 4.1: Descriptive statistics of the number of sentences per summary in the evaluation benchmark.

4.1.2 Annotation Process

The annotation process was carried out in two phases. In the first phase, the dataset instances were independently annotated by a human and two LLM-based annotators including GPT-4o-mini [34] and Gemini-Flash-2.5 [35]. In addition, we applied majority voting on the first 100 rows to aggregate the

stance annotations. For Hebrew, we saw that GPT-generated annotations have the highest agreement with the majority, therefore they were selected as the annotations for the rest of the data. For English, we used Gemini-Flash-2.5 [35] and claude-sonnet-4.5 [33] for annotations, and one human annotator. Then, the same majority technique was applied, and Claude annotations showed the strongest agreement; hence they were used as the initial reference annotations.

In the second phase, the selected annotations were manually reviewed and corrected when necessary. In cases of disagreement or ambiguity, human judgment was used to resolve conflicts and refine the final labels. This process allowed us to combine the scalability of LLM-assisted annotation with human validation and quality control.

4.1.3 Quality Analysis

To evaluate the reliability of the annotation process, we measured inter-annotator agreement (IAA) using Cohen’s κ , as shown in Figure 4.1. The results reveal substantial differences between the Hebrew and English benchmarks.

The results reveal clear differences between the English and Hebrew annotations. For Hebrew, both unanimous agreement among all three annotators and majority agreement—agreement between at least two of the three annotators—are consistently moderate to substantial. Pairwise agreement ranges from $\kappa = 0.537$ between the human annotator and Gemini to $\kappa = 0.866$ between GPT and the majority-vote annotations for summary

stance. Similar trends are observed for article stance, where agreement ranges from $\kappa = 0.581$ to $\kappa = 0.827$. These results indicate that the annotators generally converge on similar stance labels, suggesting that the task is relatively stable and consistently interpretable.

In contrast, the English benchmark exhibits much lower agreement. While Claude shows modest agreement with the human annotator ($\kappa = 0.085$ - 0.138), agreement involving Gemini is close to zero and even slightly negative in one case. Consequently, the majority-vote annotations align strongly with Claude ($\kappa = 0.556$ - 0.580) but much less with Gemini ($\kappa = 0.069$ - 0.316). This pattern suggests that disagreement is largely driven by Gemini assigning stance labels that differ substantially from those produced by the human annotator and Claude.

Across both benchmarks, majority voting increases consistency by reducing the influence of individual annotation decisions. However, the resulting labels should be viewed as consensus-based reference annotations rather than absolute gold-standard labels. Overall, the substantially higher agreement observed for Hebrew indicates that stance annotation is more stable in the Hebrew benchmark than in the English benchmark, whereas the English benchmark exhibits greater ambiguity and annotator disagreement.

Chapter 5

Experiments

5.1 Task Description

Given a source document and a generated summary, STANCESUM evaluates whether the summary preserves the stance expressed in the source document. We formulate this as a sentence-level comparison between summary sentences and their aligned source sentences.

Formally, let $D = \{d_1, \dots, d_m\}$ denote the set of source document sentences, where d_j is the j -th sentence of the document. Let $S = \{s_1, \dots, s_n\}$ denote the set of summary sentences, where s_i is the i -th sentence of the summary. For each summary sentence s_i , we denote its aligned source sentence as $d_{a(i)} \in D$, where $a(i)$ is the index of the source sentence most directly associated with s_i .

Let $Tp(x)$ denote the main topic of sentence x , and let $St(x)$ denote the stance expressed by x toward that topic. We consider an aligned pair $(s_i, d_{a(i)})$ to

be *comparable* if both sentences refer to the same topic:

$$Tp(s_i) = Tp(d_{a(i)}).$$

For comparable pairs, stance is preserved if the summary sentence and its aligned source sentence express the same stance toward that shared topic:

$$St(s_i) = St(d_{a(i)}).$$

This formulation yields three sentence-pair outcomes. If the topics match and the stance labels match, the pair is counted as stance-preserving. If the topics match but the stance labels differ, the pair is counted as a stance shift. If the topics differ, the pair is treated as topic-mismatched: it remains part of the benchmark, but it is not assigned a stance-preservation or stance-shift label because the stance targets are different.

5.1.0.0.1 Reference Preservation Score To evaluate automatic stance-preservation metrics, we aggregate the sentence-level outcomes defined above into a single document-level reference score for each document-summary pair. For a document-summary pair (D, S) , let T_{total} denote the total number of aligned summary sentences. Let T_{diff} denote the number of aligned pairs whose annotated topics differ, and let T_{match} denote the number of aligned pairs whose annotated topics and stance labels both match. We define the

StanceSum reference score as:

$$\text{Score}_{\text{ref}}(D, S) = \frac{T_{\text{match}}}{T_{\text{total}} - T_{\text{diff}}}.$$

The denominator excludes topic-mismatched pairs because stance preservation is meaningful only when the source and summary sentences discuss the same topic. Thus, $\text{Score}_{\text{ref}}$ measures the proportion of comparable sentence pairs in which the summary preserves the source stance. This aggregation yields one reference preservation score for each document-summary pair. The sentence-level annotations therefore provide the fine-grained basis for the score, while the final metric evaluation is conducted at the document-summary level.

We evaluate whether automatic metrics can estimate the STANCESUM reference preservation score. Our experiments address three research questions: **RQ1:** Do LLM-generated summaries exhibit measurable stance shifts? **RQ2:** How well do existing automatic metrics correlate with the STANCESUM reference score? **RQ3:** How does performance differ between Hebrew and English?

We evaluate metrics in two settings:

1. **Article-level preservation:** each metric receives the source document and summary as full texts and returns one document-level score.
2. **Sentence-level preservation:** metrics are computed over aligned sentence pairs, following Section 3.1.2, and then aggregated into one document-level score.

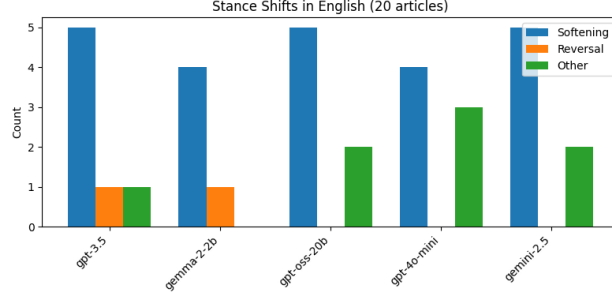
In both settings, scores are correlated with a corresponding STANCESUM reference score.

5.2 Empirical Evidence of Stance Shifts

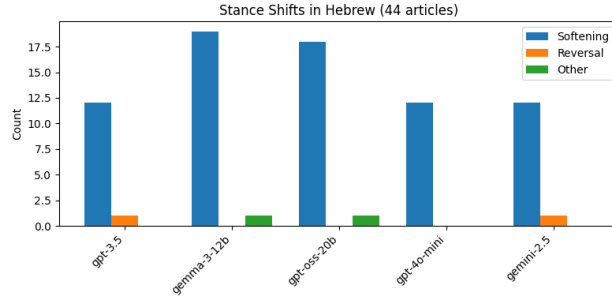
Before evaluating automatic metrics, we first conduct a diagnostic analysis addressing RQ1: whether LLM-generated summaries exhibit stance shifts. This analysis is distinct from the 50-pair STANCESUM benchmark used for metric evaluation. For Hebrew, we use 44 source documents and generate summaries for each document using five LLMs, yielding 220 generated summaries. For English, we use 20 source documents and generate summaries for each document using five LLMs, yielding 100 generated summaries.

Each generated summary is manually inspected to determine whether it contains a stance shift relative to the source document. When a shift is identified, we categorize its direction as *softening*, *reversal*, or *other*. This analysis is intended as empirical motivation rather than a large-scale estimate of stance-shift prevalence.

The results show repeated stance shifts across the evaluated models and languages as detailed in Figure 5.1. Most observed shifts correspond to stance softening, such as *Favor* $\rightarrow$ *Neutral* or *Against* $\rightarrow$ *Neutral*. Direct stance reversals are less frequent, but they are still observed. Because this analysis is diagnostic and relatively small in scale, we use it to motivate the need for stance-preservation evaluation rather than to estimate the overall prevalence of stance shifts in LLM summarization.



(a) English (20 articles)



(b) Hebrew (44 articles)

Figure 5.1: Aggregated stance shifts between source articles and generated summaries across models. Softening denotes shifts toward a neutral stance (e.g., *Favor* $\rightarrow$ *Neutral*, *Against* $\rightarrow$ *Neutral*), Reversal denotes direct polarity flips (e.g., *Favor* $\rightarrow$ *Against*, *Against* $\rightarrow$ *Favor*), and Other includes shifts originating from a neutral stance (e.g., *Neutral* $\rightarrow$ *Favor*, *Neutral* $\rightarrow$ *Against*). Across both languages, softening is the most prevalent type of shift, indicating a systematic tendency of models to neutralize stance in generated summaries.

5.3 Baselines

We use four kinds of baselines: **Lexical Similarity Baselines**, we compute four widely used text similarity metrics, BLEU, ROUGE-1, ROUGE-2, ROUGE-L. **Similarity-based baselines**, we compute cosine similarity between each source article and its corresponding summary using TF-IDF representations based on character 3-5-grams. Character-level n-grams pro-

Model	Hebrew (44 articles)						English (20 articles)					
	$F \rightarrow N$	$F \rightarrow A$	$A \rightarrow N$	$A \rightarrow F$	$N \rightarrow A$	$N \rightarrow F$	$F \rightarrow N$	$F \rightarrow A$	$A \rightarrow N$	$A \rightarrow F$	$N \rightarrow A$	$N \rightarrow F$
gpt-3.5-turbo	7	0	5	1	0	0	2	0	3	1	1	0
gemma-3-12b-it	15	0	4	0	1	0	–	–	–	–	–	–
gemma-2-2b-it	–	–	–	–	–	–	0	0	4	1	0	0
gpt-oss-20b	14	0	4	0	1	0	1	0	4	0	0	2
gpt-4o-mini	10	0	2	0	0	0	1	0	3	0	1	2
gemini-2.5-flash	10	0	2	1	0	0	1	0	4	0	0	2

Table 5.1: Counts of stance shifts between source articles and generated summaries in Hebrew and English evaluation sets. F , A , and N denote Favor, Against, and Neutral respectively.

vide a more morphology-robust lexical representation for Hebrew, since related inflected or affixed word forms share overlapping subword character patterns. In addition, we compute embedding-based cosine similarity using *text-embedding-3-small* at both the document level - between the article and summary embeddings, and the sentence level - between aligned article-summary sentence pairs. **LLM-Based Direct Scoring Baseline**, to assess whether an LLM can directly evaluate stance preservation without task-specific scaffolding, we use the OpenAI API model `gpt-4o-mini` to assign a stance-preservation quality score from 1 to 9. We normalize the resulting scores by dividing by 9 and use them as a direct LLM-based evaluation baseline. In our experiments, the model is queried with temperature 0.5. The prompts are provided in Appendix C. **Oracle-Topic Zero-Shot NLI Baseline**, we evaluate a zero-shot NLI-based stance baseline using the `joeddav/xlm-roberta-large-xnli` model [36]. For each source-summary unit, the model estimates stance toward the corresponding topic using the candidate labels *Favor*, *Against*, and *Neutral*. Given a text x and topic t , the model predicts a probability distribution over stance labels using a zero-shot

classification formulation with a hypothesis template of the form:

"This text expresses a {label} stance toward {t}."

The labels are mapped to numerical values $+1$, -1 , and 0 , respectively, and the model’s label probabilities are converted into an expected stance value.

This baseline is evaluated in an oracle-topic setting: unlike a fully standalone evaluator, it is given the topic associated with each instance rather than identifying the topic automatically. We then convert the expected stance difference between the source and summary into a preservation score:

$$\text{Score}_{\text{NLI}} = 1 - \frac{|E[\text{St}_{\text{sum}}] - E[\text{St}_{\text{src}}]|}{2}.$$

Since the maximum possible distance between expected stance values is 2, this score lies in $[0, 1]$, where higher values indicate smaller predicted stance shifts. This baseline also serves as a proxy for prior perspective-preservation evaluation settings that rely on topic-conditioned stance estimation. A direct comparison to P3SUM is not possible in our setting, since P3SUM is designed for political summaries with predefined stance targets, while our benchmark covers Hebrew and English summaries with open-ended topics that must be detected automatically.

5.4 Evaluation Metrics

To evaluate automatic stance-preservation metrics, we measure the agreement between each metric’s predicted score and the STANCESUM reference preservation score $\text{Score}_{\text{ref}}$. We report three complementary correlation measures. Pearson’s correlation coefficient (r) [37] measures the strength of the linear relationship between predicted and reference scores. Spearman’s rank correlation coefficient (ρ) [38] evaluates how well the ranking induced by the predicted scores matches the ranking of the reference scores, regardless of the exact numerical values. Kendall’s rank correlation coefficient (τ) [39] also measures rank agreement but is based on the proportion of concordant and discordant pairs, making it particularly robust for comparing ranked data. Together, these three metrics provide a comprehensive evaluation by assessing both linear agreement and ranking consistency.

To evaluate the robustness of the results, we additionally perform a grouped five-fold evaluation, where all summaries originating from the same source article are assigned to the same fold. For each fold, we compute the three correlation coefficients for every evaluated method. The resulting fold-wise correlation values are ranked and compared using the Friedman test to determine whether significant differences exist among the evaluated methods. When the Friedman test indicates statistical significance, we perform Dunn’s post-hoc test to conduct pairwise comparisons between methods. This evaluation protocol is applied independently to the English and Hebrew benchmarks at both the article and sentence levels.

Chapter 6

Results

Tables 6.2 and 6.3 report correlations between automatic metric scores and the STANCESUM reference preservation score. Having addressed RQ1 in Section 5.2, we now evaluate existing metrics, cross-lingual differences, and our stance-aware variant.

6.1 Model Development Process

The final STANCESUM framework was developed through a sequence of design experiments aimed at identifying suitable methods for each stage of the pipeline. Rather than selecting components a priori, we evaluated several alternatives for sentence alignment and stance detection before arriving at the final architecture presented in Chapter 3. This section summarizes the main development stages and the rationale behind the final design.

Initially, we envisioned the framework as a three-stage pipeline:

1. Sentence alignment between the source document and the generated summary.
2. Stance prediction for each aligned sentence pair.
3. Aggregation of the sentence-level stance predictions into a document-level stance-preservation score using KL divergence.

6.1.1 Sentence Alignment

The first design question concerned sentence alignment. Inspired by the sentence-level alignment strategy employed by SummaC, we initially investigated whether a Hebrew Natural Language Inference (NLI) model trained on the HebNLI dataset [40] could be used to identify corresponding document-summary sentence pairs. The idea was to construct an $M \times N$ entailment matrix, where each entry represented the entailment score between a source sentence and a summary sentence. For every summary sentence, the source sentence with the highest entailment score would be selected as its aligned counterpart.

In practice, this approach did not produce sufficiently reliable alignments. Although NLI models estimate whether one sentence semantically entails another, they are not optimized for retrieving the most semantically similar sentence from an entire document. As a result, many summary sentences were aligned with sentences that were logically related but not the most appropriate semantic match, reducing the reliability of the subsequent stance comparison.

We therefore reformulated sentence alignment as a semantic retrieval problem and compared three retrieval methods: TF-IDF, BM25, and multilingual-e5 (mE5). To compare the retrieval methods, we manually created seven sentence pairs expressing the same meaning using different wording. Each pair was inserted into documents from the HeSum dataset, and each retrieval method was evaluated on its ability to retrieve the corresponding sentence. Although the evaluation set was intentionally small, the performance differences between the methods were sufficiently pronounced to clearly distinguish their retrieval capabilities.

For example, for the pair:

”My dog is very cute.”

”I have a sweet dog.”

the similarity scores were:

- TF-IDF: 0.000
- BM25: 0.000
- mE5: 0.962

Similar behavior was observed across the remaining sentence pairs. TF-IDF and BM25 rely primarily on lexical overlap and therefore struggled when semantically equivalent sentences used different vocabulary. In contrast, mE5 consistently retrieved the correct sentence because dense embedding models capture semantic similarity beyond exact word matching. Based on these

observations, mE5 was selected as the sentence alignment component of the final framework.

6.1.2 Stance Detection

After selecting the sentence alignment method, we focused on the second stage of the pipeline: stance prediction. Our initial idea was to adapt an existing Hebrew sentiment analysis model. However, as discussed in Chapter 2, sentiment analysis and stance detection are fundamentally different tasks. Sentiment describes the emotional polarity of a sentence, whereas stance depends on the author’s position toward a specific target. Consequently, sentiment labels alone cannot reliably determine stance preservation.

We therefore began developing a dedicated Hebrew stance detection model, as described in Appendix A. Our first attempt used the synthetic dataset described in A.1.1.1. We fine-tuned the Hebrew sentiment model on this dataset under the assumption that its existing language representations would transfer effectively to stance detection. However, this approach produced poor results. The primary limitation was that the model attempted to predict stance without explicitly considering the discussion topic. Since stance is inherently target-dependent, identical sentences may express different stances depending on the topic being discussed. Consequently, the model was unable to generalize reliably and was not pursued further.

To address this limitation, we designed a second topic-aware stance detection model. Approximately 50 discussion topics were extracted from 15 HeSum documents using ChatGPT. For each topic, approximately ten synthetic ex-

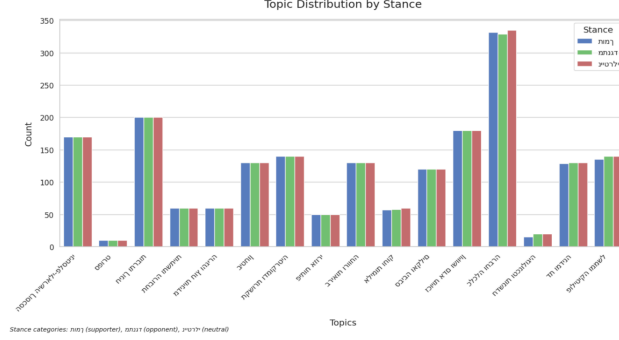
amples were generated for each stance category (Favor, Against, and Neutral). Closely related topics were subsequently merged into broader semantic categories. For example, the Hebrew topics "חינוך דתי", "הדתה" and "חופש דת" were merged into the broader category "דת ומדינה". Each training instance was then represented as a topic-stance combination (e.g., "דת" "ומדינה_תומך"). The resulting synthetic corpus contained approximately 7,000 labeled instances.

Because the generated dataset exhibited substantial class imbalance, we evaluated three fine-tuning strategies based on the `dicta-il/dictabert-sentiment` model:

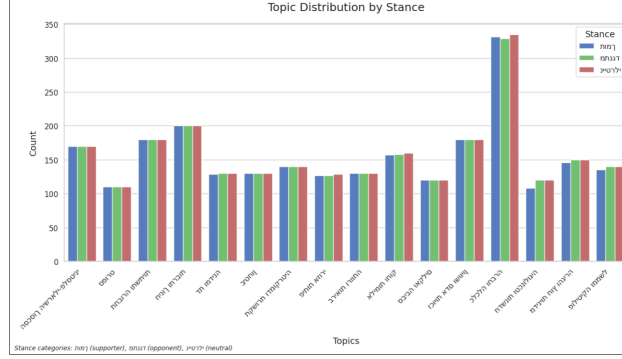
1. Weighted-loss fine-tuning, assigning larger class weights to the minority classes.
2. Data augmentation through additional synthetic examples for the minority classes.
3. A combination of weighted-loss training and data augmentation.

Figure 6.1 illustrates the class distribution before and after balancing the synthetic dataset.

The quantitative results are presented in Table 6.1. All three training strategies achieved relatively high performance on the synthetic validation and test sets. Among them, weighted-loss fine-tuning produced the best overall results, achieving an F1-score of 0.769 on the test set and stable five-fold cross-validation performance.



(a) Original class distribution



(b) After synthetic balancing

Figure 6.1: Class distribution of the synthetic topic-aware stance dataset before and after balancing.

Despite these encouraging results, the model performed considerably worse after being integrated into the complete stance-preservation pipeline. The primary reason was that the synthetic validation and test sets shared essentially the same topic distribution as the training data. As a result, the model learned topic-specific associations rather than learning to infer stance for previously unseen topics. When evaluated on real document-summary pairs containing new discussion topics, performance deteriorated substantially.

This experiment highlighted an important limitation of treating the topic

Validation	Accuracy	F1 macro	Precision	Recall
Weighted	0.8339	0.8408	0.8531	0.8434
Balanced	0.8154	0.8124	0.8206	0.8202
Combined	0.7104	0.7041	0.7238	0.7147
Test				
Weighted	0.7923	0.7694	0.7755	0.7925
Balanced	0.7599	0.7598	0.7848	0.7671
Combined	0.7525	0.7364	0.7635	0.7538
Cross Validation				
Weighted (best model)	0.9133 $\pm$ 0.0506	0.9131 $\pm$ 0.0568	0.9166 $\pm$ 0.0594	0.9150 $\pm$ 0.0513

Table 6.1: Topic in class trial.

as part of the prediction label. Since the number of possible topic-stance combinations continually grows as new topics appear, such a formulation generalizes poorly to open-domain summarization, where previously unseen topics are expected.

These findings motivated the development of the final stance detection model presented in Appendix A. Rather than treating each topic-stance combination as a separate class, the final model was designed to better generalize to unseen topics while maintaining accurate stance prediction. This final architecture was ultimately adopted in the STANCESUM evaluation framework.

6.2 Existing automatic metrics

To address RQ2, we evaluate whether existing automatic evaluation metrics are able to capture stance preservation between source documents and generated summaries.

In English, several lexical and similarity-based metrics show moderate positive correlations with the reference stance-preservation score, particularly under mean sentence-level aggregation. Among the generic baselines, LLM-

based scoring achieves the strongest article-level rank correlations ($\rho = 0.402$, $\tau = 0.343$), while ROUGE-based metrics and BLEU obtain moderate sentence-level correlations. However, these trends are inconsistent across languages. In Hebrew, most generic baselines remain relatively weak, with only modest positive correlations for BLEU, ROUGE, TF-IDF, and embedding similarity metrics.

Importantly, most baseline metrics are similarity-based measures, where higher values indicate stronger preservation and therefore are expected to correlate positively with the reference score. In contrast, our method is distance-based: lower Earth Mover’s Distance (EMD) values correspond to better stance preservation. Consequently, stronger performance for our method is reflected by larger negative correlations.

The external stance/NLI-based baseline also performs inconsistently across both languages. At the article level, its correlations remain weak or negative in English ($\rho = -0.003$) and Hebrew ($\rho = -0.199$), while sentence-level correlations are similarly unstable. These findings suggest that standard lexical similarity metrics and generic zero-shot inference approaches do not reliably capture stance preservation, particularly in multilingual and open-domain summarization settings. In contrast, our stance-preservation metric achieves the strongest overall correlations with the reference scores across both English and Hebrew, despite operating under the inverse correlation interpretation induced by the distance-based formulation.

Model	Article-Level			Mean Sentence Article Score		
	Pearson	Spearman	Kendall	Pearson	Spearman	Kendall
BLEU	0.154 [†]	0.161 [†]	0.135 [†]	0.133 [†]	0.243 [†]	0.185 [†]
ROUGE-1	0.104 [†]	0.077	0.068	0.122 [†]	0.148 [†]	0.126 [†]
ROUGE-2	0.105 [†]	0.101 [†]	0.070	0.109 [†]	0.166 [†]	0.142 [†]
ROUGE-L	0.109 [†]	0.108 [†]	0.097 [†]	0.132 [†]	0.148 [†]	0.126 [†]
TF-IDF	0.245 [†]	0.143 [†]	0.115 [†]	0.124 [†]	0.055	0.044
EMB	0.265 [†]	0.125 [†]	0.097 [†]	0.205	0.098	0.079
LLM	0.148 [†]	0.236 [†]	0.205 [†]	0.125 [†]	0.129 [†]	0.104 [†]
Zero-shot NLI	-0.081	-0.199 [†]	-0.140 [†]	0.098	0.141	0.103
Ours	-0.471[†]	-0.405[†]	-0.297[†]	-0.471[†]	-0.405[†]	-0.297[†]

Table 6.2: **Hebrew results.** Correlation between automatic metrics and the STANCESUM reference preservation score at the article level and the mean sentence article score. Highest correlations in each column are shown in bold. [†] indicates $p < 0.05$.

6.2.1 Statistical Significance Analysis

To verify that the observed performance differences are not due to random variation, we perform a statistical significance analysis over the fold-wise correlation results. For each experimental setting (English article-level, English sentence-level, Hebrew article-level, and Hebrew sentence-level), the dataset is independently partitioned into five folds using grouped cross-validation, ensuring that all summaries originating from the same source article remain within the same fold. For every fold, we compute Pearson, Spearman, and Kendall correlation coefficients between each automatic evaluation metric and the reference stance-preservation scores. The resulting fold-wise correlation values are ranked to compare the relative performance of the evaluated methods. We first apply the Friedman test to determine whether statistically significant differences exist among the evaluated methods. When the null hypothesis is rejected, Dunn’s post-hoc test is used to perform pairwise

Model	Article-Level			Mean Sentence Article Score		
	Pearson	Spearman	Kendall	Pearson	Spearman	Kendall
BLEU	0.076	-0.015	0.008	0.150 [†]	0.249 [†]	0.186 [†]
ROUGE-1	-0.044	-0.028	-0.009	0.204 [†]	0.226 [†]	0.164 [†]
ROUGE-2	0.107 [†]	0.108 [†]	0.091 [†]	0.198 [†]	0.239 [†]	0.176 [†]
ROUGE-L	0.103 [†]	0.099 [†]	0.077 [†]	0.216 [†]	0.238 [†]	0.175 [†]
TF-IDF	-0.131 [†]	-0.211 [†]	-0.150 [†]	0.156 [†]	0.158 [†]	0.115 [†]
EMB	-0.043	-0.007	0.000	0.111 [†]	0.141 [†]	0.103 [†]
LLM	0.237 [†]	0.402 [†]	0.343 [†]	0.219 [†]	0.236 [†]	0.180 [†]
Zero-shot NLI	-0.048	-0.003	-0.006	-0.098	-0.121	-0.095
Ours	-0.400[†]	-0.514[†]	-0.363[†]	-0.400[†]	-0.514[†]	-0.363[†]

Table 6.3: **English results.** Correlation between automatic metrics and the STANCESUM reference preservation score at the article level and the mean sentence article score. Highest correlations in each column are shown in bold. [†] indicates $p < 0.05$.

comparisons between methods.

Table 6.4 summarizes the rank of each evaluation method in the four experimental settings, together with its mean rank across all settings. Ranks are computed from the fold-wise correlation values, where rank 1 corresponds to the best-performing method for a given setting. The final mean rank is computed by averaging the ranks over all four settings.

Method	Article-level		Sentence-level		Mean Rank
	Hebrew	English	Hebrew	English	
STANCESUM	1	1	1	1	1
ROUGE-L	2	5	4	8	4.5
ROUGE-1	4	8	5	2	4.75
BLEU	6	3	9	5	4.75
Embedding Similarity	5	4	6	3	5
LLM Score	3	9	8	7	5.75
ROUGE-2	7	7	2	4	5.75
TF-IDF	8	2	7	6	6.75
Zero-shot NLI	9	6	3	9	6.75

Table 6.4: Average ranking of the evaluated methods based on the fold-wise correlation results. Lower ranks indicate better performance.

As shown in Table 6.5, the Friedman test rejects the null hypothesis in all four experimental settings ($p < 0.001$), indicating statistically significant

Setting	Friedman χ^2	p-value
Hebrew Article	28.7928	0.0003
Hebrew Sentence	27.7977	0.0005
English Article	55.5645	3.42e-09
English Sentence	40.2274	2.90e-06

Table 6.5: Friedman test statistics for each experimental setting.

differences among the evaluated methods.

Dunn’s post-hoc test was then applied to perform pairwise comparisons between methods. Across the four experimental settings, STANCESUM consistently achieves the highest rank among all evaluated methods (Table 6.4). Pairwise analysis further shows that STANCESUM significantly outperforms most lexical, embedding-based, LLM-based, and external evaluation baselines ($p < 0.05$ in the majority of comparisons), with only a small number of pairwise differences not reaching statistical significance.

6.3 English–Hebrew Cross-lingual differences

To address RQ3, we analyze how stance-preservation evaluation differs across English and Hebrew.

The results indicate a clear cross-lingual difference between English and Hebrew. In English, several lexical and similarity-based metrics achieve moderate correlations with the reference stance-preservation score, particularly at the mean sentence level. BLEU and ROUGE-based metrics obtain positive and statistically significant sentence-level correlations, with ROUGE-L achieving the strongest Pearson correlation among the lexical baselines ($r = 0.216$). At the article level, LLM-based scoring achieves the strongest

positive rank correlations among the generic baselines ($\rho = 0.402$, $\tau = 0.343$), while most lexical metrics remain relatively weak. Our stance-aware metric achieves the strongest overall correlations across all English evaluations, with statistically significant negative correlations due to its distance-based formulation ($r = -0.400$, $\rho = -0.514$, $\tau = -0.363$).

In contrast, the Hebrew results are substantially weaker and less consistent across the generic baselines. Although several metrics obtain small positive article-level correlations, including TF-IDF ($r = 0.245$) and embedding similarity ($r = 0.265$), their sentence-level performance remains limited and often statistically insignificant. ROUGE- and BLEU-based metrics achieve only modest positive correlations overall, while the external NLI-based baseline exhibits unstable behavior and negative article-level rank correlations ($\rho = -0.199$, $\tau = -0.140$).

These findings suggest that lexical overlap and general semantic similarity provide substantially weaker signals for stance preservation in Hebrew, likely due to the language’s richer morphology and higher lexical variation. In contrast, our stance-aware metric achieves the strongest overall correlations in Hebrew across all evaluation settings ($r = -0.471$, $\rho = -0.405$, $\tau = -0.297$), despite the inverse interpretation induced by the distance-based formulation.

Overall, our results do not suggest that generic metrics are ineffective, nor that our framework fully solves stance-preservation evaluation. Rather, they indicate that stance preservation is not reliably captured by common off-the-shelf metrics across languages, and that explicitly modeling topic-conditioned

stance provides a useful starting point for this evaluation problem.

Chapter 7

Discussion and Conclusions

7.1 Discussion

The central finding of this thesis is that modern summarization systems do not consistently preserve the stance expressed in the original source documents. Across both the Hebrew and English benchmarks, we observed cases in which generated summaries remained factually consistent and semantically faithful to the source while nevertheless altering the author’s position toward the discussed topic. The most common pattern was a shift from non-neutral stances toward a more neutral presentation, indicating that stance changes can occur even when conventional quality measures suggest that the summary is accurate.

This observation has important implications for summarization evaluation. Existing metrics primarily assess lexical overlap, semantic similarity, or factual consistency, but our results demonstrate that these properties alone are

insufficient for determining whether a summary faithfully preserves the author’s viewpoint. Stance preservation therefore emerges as a complementary evaluation dimension rather than an implicit consequence of existing quality measures.

Motivated by this finding, we developed a multilingual stance-preservation evaluation framework that measures whether the stance expressed in the source document is maintained in the generated summary. The proposed framework combines sentence-level semantic alignment, topic-aware filtering, and stance prediction to automatically quantify stance preservation. Our experimental results show that this approach captures changes in author perspective that are overlooked by existing summarization evaluation metrics, providing a practical first step toward systematic stance-aware evaluation.

7.1.1 Stance is a distinct evaluation dimension

A central motivation of this work was the observation that preserving information does not necessarily imply preserving meaning. In many real-world applications, the author’s position toward a topic forms an essential part of the communicated message. A news article criticizing a public policy, a scientific paper recommending a medical intervention, or a public-health document encouraging vaccination may remain factually accurate after summarization while nevertheless conveying a substantially different message if the expressed stance is weakened, strengthened, or reversed.

Our diagnostic analysis provides empirical evidence supporting this concern. Across multiple summarization models and across both languages, the dom-

inant error pattern was not direct polarity reversal but rather systematic softening of non-neutral stances toward a neutral position. This behavior suggests that current summarization systems may have a tendency to reduce opinionated or controversial language during summarization. While such behavior may produce summaries that appear balanced or less polarized, it may also unintentionally alter the author’s intended perspective.

More broadly, these findings reinforce the distinction between factual consistency and stance preservation. A summary may accurately report the facts contained in the original document while simultaneously changing how those facts are presented or interpreted. Consequently, evaluating summarization quality solely through factual consistency or semantic similarity may provide an incomplete picture of summary faithfulness.

7.1.2 Why current metrics are insufficient

Our experiments further demonstrate that commonly used automatic evaluation metrics are only partially sensitive to stance preservation. Lexical-overlap measures such as BLEU and ROUGE, semantic similarity metrics, and even direct LLM-based scoring exhibit only moderate correlations with the reference stance-preservation scores, particularly in English, and substantially weaker and less consistent behavior in Hebrew.

These observations are consistent with the objectives of these metrics. ROUGE and BLEU were designed to measure lexical similarity, embedding-based methods quantify semantic proximity, and factual consistency metrics estimate whether information is supported by the source document. None of

these methods explicitly models the author’s attitude toward the discussed topic. As a result, summaries that preserve content while modifying stance may still receive high scores according to existing evaluation protocols.

In contrast, explicitly modeling topic-conditioned stance distributions provides a stronger signal for detecting stance shifts. Although the proposed framework does not completely solve stance-preservation evaluation, the results indicate that incorporating topic identification together with stance prediction captures aspects of summary quality that are largely invisible to conventional evaluation metrics.

More generally, these findings suggest that summarization evaluation should increasingly be viewed as a multidimensional problem. Future evaluation frameworks may benefit from combining complementary dimensions such as factual consistency, semantic similarity, fluency, coherence, and stance preservation, rather than relying on any single automatic metric.

7.1.3 Implications for multilingual summarization

An additional contribution of this work is the multilingual evaluation of stance preservation across Hebrew and English. While several existing evaluation metrics achieved moderate performance in English, their behavior became substantially weaker and less consistent in Hebrew. This observation highlights the difficulty of transferring evaluation methods that were primarily developed and validated on English benchmarks to morphologically richer and lower-resource languages.

The stronger relative performance of the proposed stance-aware framework across both languages suggests that explicitly modeling stance may provide a more language-independent evaluation signal than relying primarily on lexical overlap. Although additional experiments on more languages are required before drawing broader conclusions, these findings indicate that multilingual summarization evaluation should consider linguistic diversity during both benchmark construction and metric design.

Finally, this work demonstrates that multilingual stance-preservation evaluation is feasible using a unified evaluation pipeline. By combining semantic alignment, topic identification, stance prediction, and distributional comparison within a single framework, STANCESUM provides an initial foundation for studying stance preservation beyond English-only settings.

7.2 Conclusion

This thesis introduced STANCESUM, a multilingual benchmark and evaluation framework for studying stance preservation in English and Hebrew summarization. Through a diagnostic analysis of multiple LLM-based summarization systems, we demonstrated that stance shifts occur across models and languages, most commonly through the softening of non-neutral positions toward a neutral stance.

Our experiments further showed that existing lexical-overlap, semantic-similarity, and general-purpose evaluation metrics only partially capture stance preservation. In contrast, explicitly modeling topic-conditioned stance distribu-

tions provides a stronger signal for evaluating whether summaries preserve the author’s original perspective.

Taken together, these findings suggest that stance preservation represents an evaluation dimension that complements factual consistency and semantic similarity rather than replacing them. As summarization systems continue to improve, evaluating whether they preserve not only information but also the author’s intended viewpoint will become increasingly important.

Beyond the proposed evaluation framework itself, this work contributes new multilingual resources, including a bilingual stance-preservation benchmark, sentence-level stance annotations, and a Hebrew stance detection model. We hope these resources will facilitate future research on stance-aware summarization evaluation and contribute toward the development of summarization systems that better preserve the meaning and perspective of the original text.

7.3 Limitations and Future Work

STANCESUM is intended as an initial diagnostic benchmark rather than a comprehensive summarization leaderboard. Because the benchmark prioritizes fine-grained sentence-level topic and stance annotation, it is currently limited in scale. The English portion is translated from the Hebrew source-summary pairs, which enables controlled Hebrew–English comparison over matched content but should be complemented in future work with naturally occurring English summarization data.

The current benchmark is also based on one Hebrew news source, and stance

annotation is inherently subjective. We therefore treat the labels as aggregated reference judgments rather than absolute ground truth. Finally, our framework uses single best-match sentence alignment and sentence-level topic matching, which may not capture one-to-many, many-to-one, or discourse-level stance relations.

Despite these limitations, we believe that STANCESUM provides a useful foundation for future research on stance-preserving summarization. Several promising research directions naturally emerge from this work.

First, the benchmark itself can be substantially expanded. Future versions of STANCESUM should include additional document collections, naturally occurring English summarization datasets, and a broader range of languages. Extending the benchmark beyond news articles to domains such as scientific publications, legal documents, healthcare communication, policy reports, and social media would enable a more comprehensive understanding of how stance preservation varies across writing styles, discourse structures, and application domains. Such an expansion would also provide a more diverse evaluation benchmark for multilingual summarization systems.

Second, the annotation methodology can be further refined. Although our benchmark combines LLM-assisted annotation with manual verification, stance annotation remains inherently subjective. Future work could incorporate multiple expert annotators, explicitly model annotator disagreement, and investigate probabilistic or confidence-weighted stance labels rather than relying on a single consensus annotation. Such approaches could better capture ambiguity in naturally occurring language and provide richer supervision for

future evaluation methods.

Another promising direction concerns the evaluation pipeline itself. Our framework currently relies on single best-match sentence alignment followed by topic filtering before comparing stance distributions. While this design provides a practical and interpretable solution, future work could investigate richer alignment strategies, including many-to-many sentence alignment, paragraph-level alignment, discourse-aware alignment, and event-based matching. Similarly, improvements in automatic topic identification, particularly for implicit or overlapping topics, could further improve the robustness of stance-preservation evaluation.

Future research may also explore richer representations of stance. In this work, stance is modeled using three discrete classes: Favor, Neutral, and Against. However, real-world discourse often contains varying degrees of certainty, hedging, mixed opinions, sarcasm, quoted viewpoints, and implicit attitudes that cannot always be represented by a small set of discrete labels. Modeling continuous or fine-grained stance representations may therefore provide a more expressive characterization of how perspectives are preserved during summarization.

Beyond improving evaluation, an important long-term direction is to develop summarization systems that explicitly preserve stance during generation. The benchmark introduced in this thesis provides a foundation for studying training objectives and decoding strategies that optimize not only factual consistency and fluency but also preservation of the author’s intended perspective. Future work may investigate reinforcement learning, reranking

methods, multi-objective optimization, or other training approaches that directly incorporate stance-preservation objectives.

Finally, an important open question concerns the effect of stance shifts on human readers. While this thesis focuses on automatic evaluation, future user studies could investigate whether readers consistently perceive stance shifts, how such shifts influence trust and perceived credibility, and whether preserving stance improves users’ understanding of the original document. Such human-centered evaluation would complement automatic metrics and provide further insight into the practical importance of stance preservation in real-world summarization systems.

Overall, we view STANCESUM as an initial step toward a broader research agenda on stance-aware summarization evaluation. As summarization systems continue to improve and become increasingly integrated into information retrieval, question answering, and decision-support applications, ensuring that generated summaries preserve not only factual information but also the author’s intended perspective will become an increasingly important challenge for the NLP community.

Appendix A

Stance Detection Model

A.1 Stance Detection Training

The Hebrew stance classifier used in our framework was trained independently from the STANCESUM evaluation benchmark. Its training data was constructed from HeSum, whereas STANCESUM was constructed from `summarization_he` and serves as the benchmark for evaluating stance-preservation metrics.

A.1.1 HeSum-Stance Training Dataset

Our primary training data for the Hebrew stance classifier is derived from the HeSum dataset [27], a corpus of long Hebrew news articles paired with human-written abstractive summaries. This training dataset is distinct from the STANCESUM benchmark, which is constructed from `summarization_he` and used for metric evaluation. To construct the sentence-level stance train-

ing data, we processed the HeSum summaries as follows.

First, each summary was segmented into individual sentences. For each sentence, we applied the dicta-il/dictalm2.0 model to detect its primary topic. Then, for every (sentence, topic) pair, we used the OpenAI API (gpt-4o-mini) to assign a stance label: Favor, Against, or Neutral.

This procedure resulted in approximately 18,000 labeled sentence level stance instances.

To validate the quality of the automatically generated labels, we manually annotated 100 randomly sampled instances to establish a ground-truth set. We then benchmarked three large language models, `gpt-4o-mini`, `claude-sonnet-3`, and `gemini-1.5-pro`, against this ground truth. Agreement with human annotations was measured using Cohen’s κ . The gpt-4o-mini model achieved the highest agreement score and was therefore selected for large scale automatic annotation.

A.1.1.1 Synthetic Stance Dataset

To improve class balance and provide controlled stance examples, we generated a supplementary synthetic dataset. We extracted topics from 15 texts in the summarization_he dataset [28]. For each topic, we prompted the OpenAI model `gpt-4o-mini` to generate 10 sentences per stance category Favor, Against, and Neutral. This process yielded a balanced synthetic dataset of approximately 5,000 instances, evenly distributed across the three stance classes. The synthetic data complements the naturally occurring news data and improves training stability.

A.1.2 Model

For Hebrew, we train a supervised stance classifier by fine-tuning DictaBERT-sentiment [41] on sentence-level data derived from the HeSum-Stance dataset. The dataset was partitioned into sentence-topic instances and labeled using the automatic annotation procedure described above, with a smaller manually annotated sample used for label-quality validation. Hyperparameters were selected using Optuna [42] on the validation set, yielding a learning rate of 1.7×10^{-5} , 3 epochs, weight decay of 0.004, and a warmup ratio of 0.144.

A.1.3 Results

We evaluate the Hebrew stance classifier in two settings. First, we use a random held-out test split of the constructed HeSum-Stance dataset. On this constructed-label test split, the model achieves an accuracy of 83.1% and a macro-F1 score of 80.9%, with consistently high AUC-ROC scores across stance categories (Table A.1). The classifier outputs probability distributions over the stance labels for each sentence, which are later used by the stance-preservation metric. The confusion matrix in Table A.2 shows balanced performance across classes, with most errors involving neutral cases, which is expected given the inherent ambiguity of neutral stance.

Second, to evaluate topic generalization, we conduct leave-one-topic-out (LOTO) experiments. In this setting, all instances from selected topic groups are excluded from training and validation and used only for testing. Performance remains fairly stable across held-out topic groups, although stance classifi-

Metric	Score
Accuracy	0.831
Macro F1	0.809
Quadratic Weighted Kappa	0.716
AUC-ROC (Favor)	0.948
AUC-ROC (Against)	0.942
AUC-ROC (Neutral)	0.931

Table A.1: Hebrew stance classifier performance on a random held-out test split of the constructed HeSum-Stance dataset.

cation for the *Ethics* topic is the most challenging. Even when excluding all three topic clusters, the model maintains an accuracy above 80%, as shown in Table A.3.

True \ Predicted	Favor	Against	Neutral
Favor	361	44	48
Against	43	596	79
Neutral	62	92	639

Table A.2: Confusion matrix for the Hebrew stance detection model. Rows correspond to true labels and columns correspond to predicted labels.

Excluded Topic(s)	Accuracy	Macro F1	F1 Loss
Freedom of speech	0.875	0.872	0.533
Security (+related)	0.782	0.773	0.677
Ethics (+related)	0.778	0.704	0.849
All three groups	0.802	0.793	1.228

Table A.3: Leave-one-topic-out evaluation for the Hebrew stance classifier. All instances from the listed topic group are excluded from training and validation and used only for testing. This evaluates generalization to unseen topics within the constructed HeSum-Stance dataset.

These evaluations are computed on held-out splits of the constructed HeSum-Stance dataset. Because the dataset labels are produced by the automatic annotation procedure described above and manually checked on a smaller validation sample, these results should be interpreted as validation on the constructed training-label distribution rather than as evaluation on a large

independently human-labeled benchmark. A larger fully human-labeled test set and calibration analysis are left for future work.

Appendix B

Topic Detection Prompt

B.0.1 Few-Shot English Prompt

Instructions:

Given a context and a sentence from it, you need to read the sentence, analyze it briefly, and then return the main topic the sentence deals with - use the context as reference.

Definitions:

The topic is the main field of the sentence (e.g., politics, medicine, education, sports, economy, security, technology, etc.).

Do not provide more than one topic.

The topic should be a single word or a short phrase (up to 3 words).

There is no need for phrases like "the topic is" - just write the topic.

If the topic cannot be identified - write: unclear.

When there is an acting entity (e.g., "Israel announced that..."), identify the field the statement deals with, not always the name of the

acting body.

There are cases where you need to be more general or more specific, depending on the meaning of the sentence.

If the sentence deals with several topics, choose the most central topic.

If the topic you found is plural - convert it to singular.

Work Steps:

1. Analyze the meaning of the sentence.
2. Identify which field the sentence deals with.
3. Return the topic.

Context: {context}

Examples:

sentence: It emerges from an analysis of 7,500 studies published on the subject between the years 1973 and 2011 that over the past 40 years there has been an ongoing downward trend in male fertility in the Western world, and a decline of more than 50 percent has been recorded in sperm concentration and sperm count.

analysis: This refers to male fertility in the Western world, and a decline in sperm concentration and sperm count.

topic: Male fertility

sentence: With the heat loads of summer and the increase in the use of air conditioners, malfunctions also increase, and this guide reviews the common malfunctions, repair costs, and recommendations for choosing an air conditioner.

analysis: This concerns air conditioners, common malfunctions, and repair costs.

topic: Air conditioning maintenance

sentence: The Ministry of Health published the coronavirus morbidity

data: the percentage of positive cases has decreased; in Israel there are 8,310 active patients; the total number of recovered stands at 325,862; the number of deaths since the outbreak of the pandemic stands at 2,735.

analysis: This concerns coronavirus morbidity data in Israel.

topic: Covid-19

sentence: As part of these creative solutions, the childhood games

of the parents' generation—who did not know the internet, tablets, iPhones, or Xbox, and made do with street games with other children—have recently been returning to us.

analysis: This refers to the childhood games of the parents' generation, games without modern technology.

topic: Games

sentence: Israel declared that it will not allow the rehabilitation of Gaza without a solution to the issue of the captives and the missing, but it remains to be seen whether it will succeed in upholding its promise.

analysis: This refers to Israel's declaration regarding the rehabilitation of Gaza and the issue of the captives and the missing.

topic: Gaza rehabilitation

sentence: Since the price between them was equalized, leisure vehicles have been nipping at the heels of family cars in sales, with importers of family cars being forced to lower prices or raise the level of equipment.

analysis: This concerns leisure vehicles versus family cars, and the

effect of price on sales.

topic: SUVs

sentence: The fungus, cordyceps by name, takes over the brains of ants and compels them to climb to great heights in order to disperse its spores.

analysis: This refers to a fungus called cordyceps that affects ants.

topic: Cordyceps

Sentence: {sentence}

Analysis:

Topic:

B.0.2 Few-Shot Hebrew Prompt

הוראות:

עליך לקרוא את המשפט הנתון, לנתח אותו בקצרה, ולאחר מכן להחזיר את הנושא המרכזי שבו המשפט עוסק.

הגדרות:

הנושא הוא התחום המרכזי של המשפט (למשל: פוליטיקה, רפואה, חינוך, ספורט, כלכלה, ביטחון, טכנולוגיה ועוד).

אל תיתן יותר מנושא אחד.

הנושא צריך להיות מילה אחת או ביטוי קצר (עד 3 מילים).

אין צורך בניסוחים כמו "הנושא הוא" – כתוב רק את הנושא.

אם לא ניתן לזהות נושא – כתוב: לא ברור.

כאשר קיימת ישות פועלת (למשל: "ישראל הודיעה כי..."), זהה את התחום שבו עוסקת ההצהרה, ולא את שם הגוף הפועל.

יש מקרים בהם תצטרך להיות כללי יותר או ספציפי יותר, בהתאם למשמעות המשפט.
אם המשפט עוסק בכמה נושאים, בחר את הנושא המרכזי ביותר.
אם הנושא שמצאת הוא ברבים – הפוך אותו ליחיד.

שלבי עבודה:

1. נתח את משמעות המשפט.
2. זהה על איזה תחום עוסק המשפט.
3. החזר את הנושא.

קונטקסט: {context}

דוגמאות:

משפט: עולה מניתוח של 7,500 מחקרים שפורסמו בנושא בין השנים 1973 עד 2011 כי ב-40 השנים האחרונות יש מגמת ירידה מתמשכת בפוריות הגבר בעולם המערבי ונרשמה ירידה של יותר מ-50 אחוזים בריכוז ובספירת הזרע.
ניתוח: מדובר על פוריות הגבר בעולם המערבי, וירידה בריכוז ובספירת הזרע.
נושא: פוריות הגבר

משפט: עם עומסי החום של הקיץ והגידול בשימוש במזגנים, עולות גם התקלות, ומדריך זה סוקר את התקלות הנפוצות, עלויות התיקון והמלצות לבחירת מזגן.
ניתוח: מדובר על מזגנים, תקלות נפוצות ועלויות תיקון.
נושא: תחזוקת מזגנים

משפט: משרד הבריאות פרסם את נתוני התחלואה בקורונה: אחוז החיוביים ירד; בישראל יש 8,310 חולים פעילים; סך המחלימים עומד על 325,862; מספר הנפטרים מפרוץ המגפה עומד על 2,735.
ניתוח: מדובר על נתוני תחלואה בקורונה בישראל.
נושא: קורונה

משפט: כחלק מהפתרונות היצירתיים הללו חוזרים אלינו לאחרונה משחקי הילדות של

דור ההורים שלא ידע אינטרנט, טאבלט, אייפון ואקס בוקס, והסתפק במשחקי רחוב

עם שאר הילדים

ניתוח: מדובר על משחקי הילדות של דור ההורים, משחקים ללא טכנולוגיה מודרנית.

נושא: משחקים

משפט: ישראל הצהירה כי לא תאפשר שיקום עזה ללא פתרון לסוגיית השבויים

והנעדרים, אך נותר לראות אם תצליח לעמוד בהבטחתה

ניתוח: מדובר על הצהרת ישראל בנוגע לשיקום עזה ולסוגיית השבויים והנעדרים.

נושא: שיקום רצועת עזה

משפט: מאז שהמחיר ביניהם השווה, רכבי הפנאי מזנבים במכירות המשפחתיים,

כשיבואני המשפחתיים נאלצים להוריד מחירים או להעלות ברמת האבזור

ניתוח: מדובר על רכבי פנאי לעומת רכבים משפחתיים, והשפעת המחיר על המכירות.

נושא: רכבי פנאי

משפט: הפטרייה, קורדיספס שמה, משתלטת על מוחן של נמלים וכופה עליהן לטפס

לגובה רב כדי לפזר את נבגיה

ניתוח: מדובר על פטרייה בשם קורדיספס שמשפיעה על נמלים.

נושא: קורד

משפט:

ניתוח:

נושא:

Appendix C

Stance Preservation Prompt

“Given the following sentence of a summary and its matching sentence from an article, rate the quality of the stance preservation in the summary on a scale from 1 to 9, where 1 is very poor and 9 is excellent. Remember that stance is not sentiment. Stance refers to the position or attitude expressed in the text towards a particular topic or entity. Take into account the relation between the summary sentence and the article sentence. If the stance is not preserved, reduce points from the score.

Article: {article}

Summary: {summary}

Provide only the numeric score. No additional text.”

Bibliography

- [1] Fabbri, Alexander R, Kryściński, Wojciech, McCann, Bryan, Xiong, Caiming, Socher, Richard, and Radev, Dragomir. Summeval: Re-evaluating summarization evaluation. *Transactions of the Association for Computational Linguistics*, 9:391–409, 2021.
- [2] Scialom, Thomas, Dray, Paul-Alexis, Lamprier, Sylvain, Piwowarski, Benjamin, Staiano, Jacopo, Wang, Alex, and Gallinari, Patrick. Questeval: Summarization asks for fact-based evaluation. In *Proceedings of the 2021 conference on empirical methods in natural language processing*, pages 6594–6604, 2021.
- [3] Wang, Alex, Cho, Kyunghyun, and Lewis, Mike. Asking and answering questions to evaluate the factual consistency of summaries. In *Proceedings of the 58th annual meeting of the association for computational linguistics*, pages 5008–5020, 2020.
- [4] Du Bois, John W. The stance triangle. In Englebretson, Robert, editor, *Stancetaking in Discourse*, pages 139–182. John Benjamins, Amsterdam, 2007.

- [5] Mohammad, Saif, Kiritchenko, Svetlana, Sobhani, Parinaz, Zhu, Xiaodan, and Cherry, Colin. Semeval-2016 task 6: Detecting stance in tweets. In *Proceedings of the 10th international workshop on semantic evaluation (SemEval-2016)*, pages 31–41, 2016.
- [6] Küçük, Dilek and Can, Fazli. Stance detection: A survey. *ACM Computing Surveys (CSUR)*, 53(1):1–37, 2020.
- [7] Lai, Mirko, Tambuscio, Marcella, Patti, Viviana, Ruffo, Giancarlo, and Rosso, Paolo. Stance polarity in political debates: A diachronic perspective of network homophily and conversations on twitter. *Data & Knowledge Engineering*, 124:101738, 2019.
- [8] Khiabani, Parisa Jamadi and Zubiaga, Arkaitz. Cross-target stance detection: A survey of techniques, datasets, and challenges. *Expert Systems with Applications*, page 127790, 2025.
- [9] Liu, Yuhan, Feng, Shangbin, Han, Xiaochuang, Balachandran, Vidhisha, Park, Chan Young, Kumar, Sachin, and Tsvetkov, Yulia. P³Sum: Preserving author’s perspective in news summarization with diffusion language models. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 2154–2173, Mexico City, Mexico, June 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.naacl-long.119. URL <https://aclanthology.org/2024.naacl-long.119/>.
- [10] Ri, Narutatsu, Deas, Nicholas, and McKeown, Kathleen. Reranking-

- based generation for unbiased perspective summarization. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 24701–24723, Vienna, Austria, July 2025. Association for Computational Linguistics. doi: 10.18653/v1/2025.findings-acl.1268. URL <https://aclanthology.org/2025.findings-acl.1268/>.
- [11] Zhou, Shengping, Lin, Junjie, Tan, Lianzhi, and Liu, Xin. Condensed convolution neural network by attention over self-attention for stance detection in twitter. In *2019 International Joint Conference on Neural Networks (IJCNN)*, pages 1–8. IEEE, 2019.
- [12] Yang, Yuanyu, Wu, Bin, Zhao, Kai, and Guo, Wenying. Tweet stance detection: A two-stage dc-bilstm model based on semantic attention. In *2020 IEEE Fifth International Conference on Data Science in Cyberspace (DSC)*, pages 22–29. IEEE, 2020.
- [13] Liu, Guangzhen, Zhao, Kai, Zhang, Linlin, Bi, Xuehua, Lv, Xiaoyi, and Chen, Cheng. A survey of zero-shot stance detection. In *CCF International Conference on Natural Language Processing and Chinese Computing*, pages 107–120. Springer, 2024.
- [14] Sepúlveda-Torres, Rodrigo et al. Exploring summarization to enhance headline stance detection. In *Proceedings of the International Conference on Applications of Natural Language to Information Systems (NLDB)*, pages 277–289. Springer, 2021. doi: 10.1007/978-3-030-80599-9_2.
- [15] Rubner, Yossi, Tomasi, Carlo, and Guibas, Leonidas J. A metric for distributions with applications to image databases. In *Proceedings of the Sixth*

- International Conference on Computer Vision (ICCV)*, pages 59–66. IEEE, 1998. doi: 10.1109/ICCV.1998.710701.
- [16] Gül, Ilker, Lebre, Rémi, and Aberer, Karl. Stance detection on social media with fine-tuned large language models. *arXiv preprint arXiv:2404.12171*, 2024.
- [17] Zhao, Chenye, Li, Yingjie, Caragea, Cornelia, and Zhang, Yue. Zerostance: Leveraging chatgpt for open-domain stance detection via dataset generation. In *Findings of the Association for Computational Linguistics ACL 2024*, pages 13390–13405, 2024.
- [18] Zhang, Bowen, Dai, Genan, Niu, Fuqiang, Yin, Nan, Fan, Xiaomao, Wang, Senzhang, Cao, Xiaochun, and Huang, Hu. A survey of stance detection on social media: New directions and perspectives. *arXiv preprint arXiv:2409.15690*, 2024.
- [19] Huang, Weidong and Yang, Jinyuan. A multi-stance detection method by fusing sentiment features. *Applied Sciences*, 14(9):3916, 2024.
- [20] Barel, Guy, Tsur, Oren, and Vilenchik, Dan. Acquired taste: Multimodal stance detection with textual and structural embeddings. *arXiv preprint arXiv:2412.03681*, 2024.
- [21] Papineni, Kishore, Roukos, Salim, Ward, Todd, and Zhu, Wei-Jing. Bleu: a method for automatic evaluation of machine translation. In *Proceedings of the 40th Annual Meeting of the Association for Computational Linguistics*, pages 311–318, Philadelphia, Pennsylvania, USA, July 2002. Association

- tion for Computational Linguistics. doi: 10.3115/1073083.1073135. URL <https://aclanthology.org/P02-1040/>.
- [22] Lin, Chin-Yew. ROUGE: A package for automatic evaluation of summaries. In *Text Summarization Branches Out*, pages 74–81, Barcelona, Spain, July 2004. Association for Computational Linguistics. URL <https://aclanthology.org/W04-1013/>.
- [23] Kryściński, Wojciech, McCann, Bryan, Xiong, Caiming, and Socher, Richard. Evaluating the factual consistency of abstractive text summarization. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 9332–9346, Online, November 2020. Association for Computational Linguistics. doi: 10.18653/v1/2020.emnlp-main.750. URL <https://aclanthology.org/2020.emnlp-main.750/>.
- [24] Laban, Philippe, Schnabel, Tobias, Bennett, Paul N., and Hearst, Marti A. SummaC: Re-visiting NLI-based models for inconsistency detection in summarization. *Transactions of the Association for Computational Linguistics*, 10:163–177, 2022. doi: 10.1162/tacl_a00453. URL <https://aclanthology.org/2022.tacl-1.10/>.
- [25] Deas, Nicholas and McKeown, Kathleen. Summarization of opinionated political documents with varied perspectives. In *Proceedings of the 31st International Conference on Computational Linguistics*, pages 8088–8108, 2025.
- [26] Liscio, Enrico, Lorandi, Michela, and Murukannaiah, Pradeep K. News is more than a collection of facts: Moral frame preserving news summarization. *arXiv preprint arXiv:2504.00657*, 2025.

- [27] Paz-Argaman, Tzuf, Mondshine, Itai, Achi Mordechai, Asaf, and Tsarfaty, Reut. HeSum: a novel dataset for abstractive text summarization in Hebrew. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 6378–6388, Bangkok, Thailand, August 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.findings-acl.381. URL <https://aclanthology.org/2024.findings-acl.381/>.
- [28] IAHLT Lab. summarization_he: Hebrew summarization dataset. https://github.com/IAHLT/summarization_he, 2023.
- [29] Chen, Yulong, Zhang, Huajian, Zhou, Yijie, Bai, Xuefeng, Wang, Yueguan, Zhong, Ming, Yan, Jianhao, Li, Yafu, Li, Judy, Zhu, Xianchao, and Zhang, Yue. Revisiting cross-lingual summarization: A corpus-based study and a new benchmark with improved annotation. In Rogers, Anna, Boyd-Graber, Jordan, and Okazaki, Naoaki, editors, *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 9332–9351, Toronto, Canada, July 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.acl-long.519. URL <https://aclanthology.org/2023.acl-long.519/>.
- [30] Wang, Liang, Yang, Nan, Huang, Xiaolong, Yang, Linjun, Majumder, Rangan, and Wei, Furu. Multilingual e5 text embeddings: A technical report. *arXiv preprint arXiv:2402.05672*, 2024.
- [31] Shmidman, Shaltiel, Shmidman, Avi, Cohen, Amir DN, and Koppel, Moshe. Adapting llms to hebrew: Unveiling dictalm 2.0

- with enhanced vocabulary and instruction capabilities, 2024. URL <https://arxiv.org/abs/2407.07080>.
- [32] Team, Gemma, Riviere, Morgane, Pathak, Shreya, Sessa, Pier Giuseppe, Hardin, Cassidy, Bhupatiraju, Surya, Hussenot, Léonard, Mesnard, Thomas, Shahriari, Bobak, Ramé, Alexandre, et al. Gemma 2: Improving open language models at a practical size. *arXiv preprint arXiv:2408.00118*, 2024.
- [33] Anthropic. Claude sonnet 4.5. <https://www.anthropic.com/claude/sonnet>, 2025. Accessed: 2026-06-17.
- [34] OpenAI. Gpt-4o mini: advancing cost-efficient intelligence. <https://openai.com/index/gpt-4o-mini-advancing-cost-efficient-intelligence/>, 2024. Accessed: 2026-06-17.
- [35] Google. Gemini 2.5 flash. <https://ai.google.dev/gemini-api/docs/models/gemini-2.5-flash>, 2025. Accessed: 2026-06-17.
- [36] Davison, Joe. joeddav/xlm-roberta-large-xnli. <https://huggingface.co/joeddav/xlm-roberta-large-xnli>, 2020. Model card for zero-shot classification model based on XLM-RoBERTa fine-tuned on multilingual NLI; accessed 2026-03-25.
- [37] Pearson, Karl and Galton, Francis. Vii. note on regression and inheritance in the case of two parents. *Proceedings of the Royal Society of London*, 58(347-352):240–242, 1895. doi: 10.1098/rspl.1895.0041. URL <https://royalsocietypublishing.org/doi/abs/10.1098/rspl.1895.0041>.

- [38] Spearman, Charles. The proof and measurement of association between two things. 1961.
- [39] Kendall, Maurice G. A new measure of rank correlation. *Biometrika*, 30 (1-2):81–93, 1938.
- [40] HebArabNlpProject. HebNLI: A natural language inference dataset in hebrew. <https://huggingface.co/datasets/HebArabNlpProject/HebNLI>, 2024. Hugging Face dataset.
- [41] Shmidman, Shaltiel, Shmidman, Avi, and Koppel, Moshe. Dictabert: A state-of-the-art bert suite for modern hebrew, 2023.
- [42] Akiba, Takuya, Sano, Shotaro, Yanase, Toshihiko, Ohta, Takeru, and Koyama, Masanori. Optuna: A next-generation hyperparameter optimization framework. In *Proceedings of the 25th ACM SIGKDD international conference on knowledge discovery & data mining*, pages 2623–2631, 2019.

תוכן העניינים

viii	רשימת איורים
x	רשימת טבלאות
1	1 מבוא
4	1.1 התרומות העקריות של המחקר
6	2 סקירת ספרות
6	1.2 זיהוי עמדה
8	2.2 זיהוי נושא או מטרה לעמדה
10	3.2 נאמנות, פרספקטיבה ועמדה בסיכום
13	4.2 סיכומים לשפות שונות ולעברית
15	5.2 מטרות המחקר
18	3 מתודולוגיה
18	1.3 סקירה כללית
20	1.1.3 פיצול משפטים
20	2.1.3 התאמת משפטים
21	3.1.3 סינון לפי נושא
21	4.1.3 זיהוי עמדה
22	5.1.3 השוואה וכינוס ציון

25 4 מערכי נתונים

25 1.4 מערך נתונים

25 1.1.4 מערך נתונים להערכה

27 2.1.4 תהליך התיוג

28 3.1.4 ניתוח איכות

30 5 ניסויים

30 1.5 תיאור משימה

33 2.5 ראייה אמפירית לשינוי עמדה

34 3.5 קווי בסיס

37 4.5 מדדי הערכה

38 6 תוצאות

38 1.6 תהליך פיתוח המודל

39 1.1.6 התאמת משפטים

41 2.1.6 זיהוי עמדה

44 2.6 מטריקות אוטומטיות קיימות

46 1.2.6 נתוח חשיבות סטטיסטית

48 3.6 הבדלים בין שפות

51 7 דיון ומסקנות

51 1.7 דיון

52	עמדה היא מימד יחודי להערכה	1.1.7
53	מדוע המדדים הנוכחיים אינם מספקים	2.1.7
54	השלכות על סיכום רב-לשוני	3.1.7
55	סיכום	2.7
56	מגבלות ועבודה עתידית	3.7
60	מודל זיהוי עמדה	נספח א
60	אימון זיהוי עמדה	1.A
60	מערך נתונים למודל זיהוי עמדה	1.1.A
61	מערך נתונים סינתטי	1.1.A
62	מודל	2.1.A
62	תוצאות	3.1.A
65	הנחיה לזיהוי נושא	נספח ב
71	הנחיה לבדיקת שימור עמדה	נספח ג
71	ביבליוגרפיה	

StanceSum: מדד חדש לסיכום שימור עמידה באנגלית

ובעברית

טליה סעדה

עבודת גמר לתואר מוסמך למדעים

אוניברסיטת בן-גוריון בנגב

תשפ"ו

תקציר

מודלי שפה גדולים נמצאים בשימוש הולך וגובר לצורכי סיכום טקסטים, אך גם כאשר הסיכומים שוטפים, רלוונטיים ועקביים עובדתית, הם עשויים לשנות את העמדה המובעת במסמך המקור. באמצעות ניתוח אמפירי של סיכומים שנוצרו על ידי מודלי שפה, אנו מראים כי מתרחשים שינויי עמדה חוזרים בין מודלים ושפות שונות, בעיקר בצורה של ריכוך עמדות לא-נייטרליות לעבר עמדה נייטרלית. לצורך חקר תופעה זו אנו מציגים את StanceSum, מערכת הערכה ומאגר נתונים רב-לשוני להערכת שימור עמדה בסיכומים באנגלית ובעברית. המערכת המוצעת מבצעת התאמה בין משפטים סמנטיים במסמך המקור ובסיכום, מסננת זוגות משפטים לפי עקביות בנושאים, ומשווה בין התפלגויות העמדה החזויות באמצעות מדדי שונות התפלגותיים. כחלק מהמחקר בנינו משאבים חדשים לעברית עבור סיכום מודע-עמדה, ובהם מאגר נתונים ברמת המשפט לזיהוי עמדה, מודל עברי לזיהוי עמדה, ובנצ'מרק דו-לשוני להערכת שימור עמדה. תוצאות הניסויים מראות כי מדדי חפיפה לקסיקלית ודמיון סמנטי משיגים מתאמים בינוניים

עם ציוני שימור עמדה באנגלית, אך ביצועיהם חלשים ופחות יציבים בעברית. לעומת זאת, המדדים ההתפלגותיים מודעי-העמדה שהצענו מציגים התנהגות יציבה יותר בין שפות ומצליחים ללכוד באופן עקבי אותות של שימור עמדה. ממצאים אלו מדגישים כי שימור עמדה מהווה ממד הערכה נפרד מעבר לעקביות עובדתית ולדמיון טקסטואלי, במיוחד בהקשרים רב-לשוניים ובתחומים שבהם לעמדת הכותב יש השפעה מהותית על פרשנות הטקסט.



אוניברסיטת בן-גוריון בנגב
הפקולטה למדעי המחשב והמידע ע"ש שטיין
המחלקה להנדסת מערכות תכנה ומידע

StanceSum: מדד חדש לסיכום שימור עמידה באנגלית ובעברית

חיבור זה מהווה חלק מהדרישות לקבלת התואר מוסמך למדעים (M.Sc)

טליה סעדה

בהנחיית פרופסור ברכה שפירא ופרופסור ליאור רוקח

_____ תאריך:	<u>Talia Seadfa</u> חתימת הסטודנט/ית:
_____ תאריך:	<u>Bracha Shapira</u> חתימת המנחה:
_____ תאריך:	<u>Rokach</u> חתימת המנחה:
_____ תאריך:	_____ חתימת יו"ר ועדת הוראה מוסמכים:

אלול תשפ"ו



אוניברסיטת בן-גוריון בנגב
הפקולטה למדעי המחשב והמידע ע"ש שטיין
המחלקה להנדסת מערכות תכנה ומידע

StanceSum: מדד חדש לסיכום שימור עמידה באנגלית ובעברית

חיבור זה מהווה חלק מהדרישות לקבלת התואר מוסמך למדעים (M.Sc)

טליה סעדה

בהנחיית פרופסור ברכה שפירא ופרופסור ליאור רוקח

אלול תשפ"ו